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CHEN et al.(10) **Pub. No.: US 2018/0348484 A1**(43) **Pub. Date: Dec. 6, 2018**(54) **IMAGE CAPTURING LENS ASSEMBLY,
IMAGING APPARATUS AND ELECTRONIC
DEVICE**(52) **U.S. Cl.**CPC *G02B 13/0045* (2013.01); *G02B 27/646*
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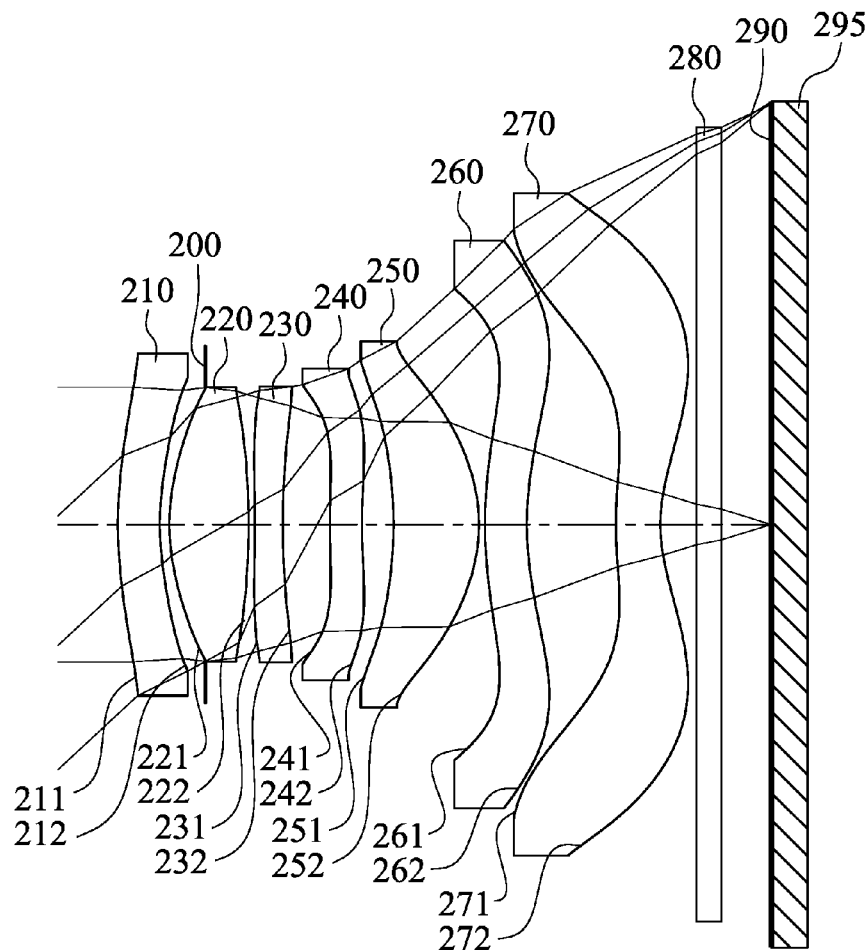
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ABSTRACT

An image capturing lens assembly includes seven lens elements, which are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element. The fifth lens element with positive refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. The sixth lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, and the image-side surface thereof includes at least one convex shape in an off-axis region thereof.



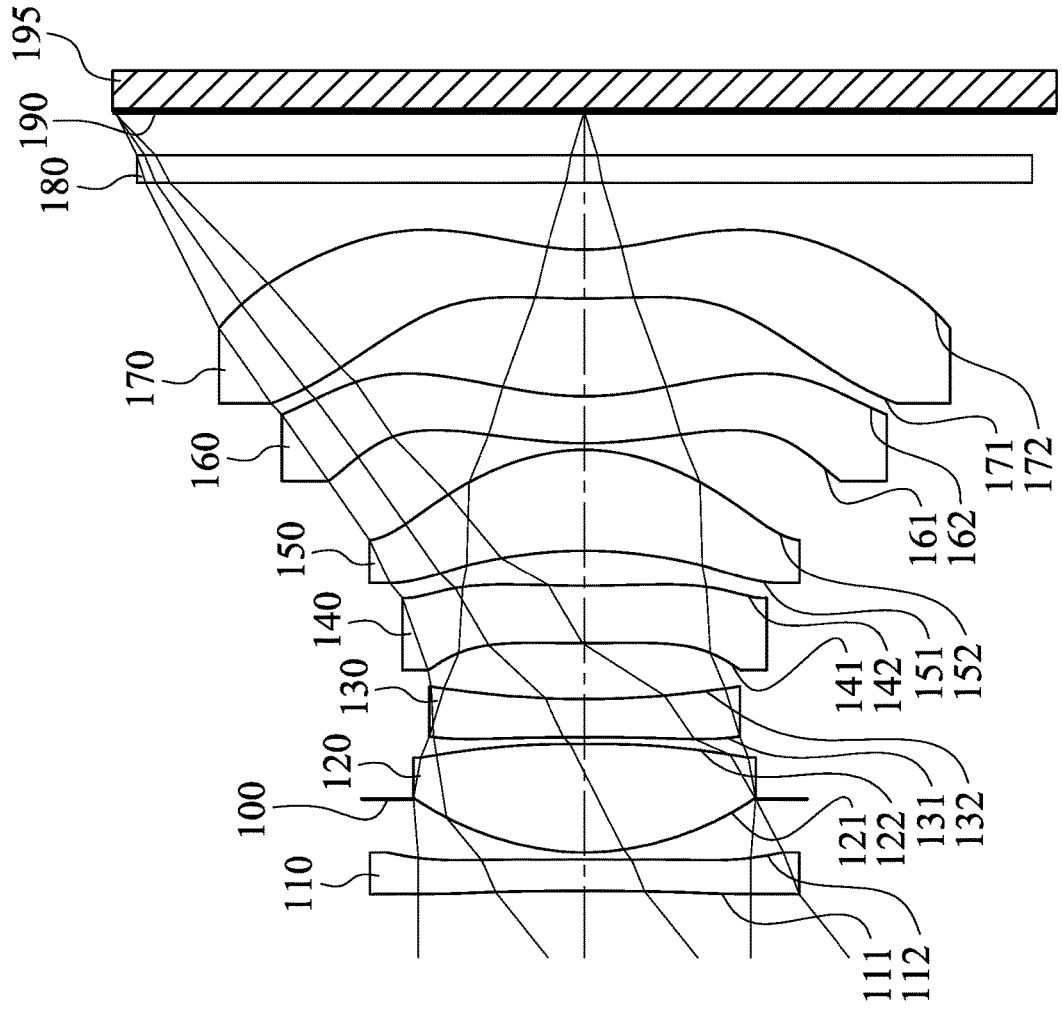


Fig. 1

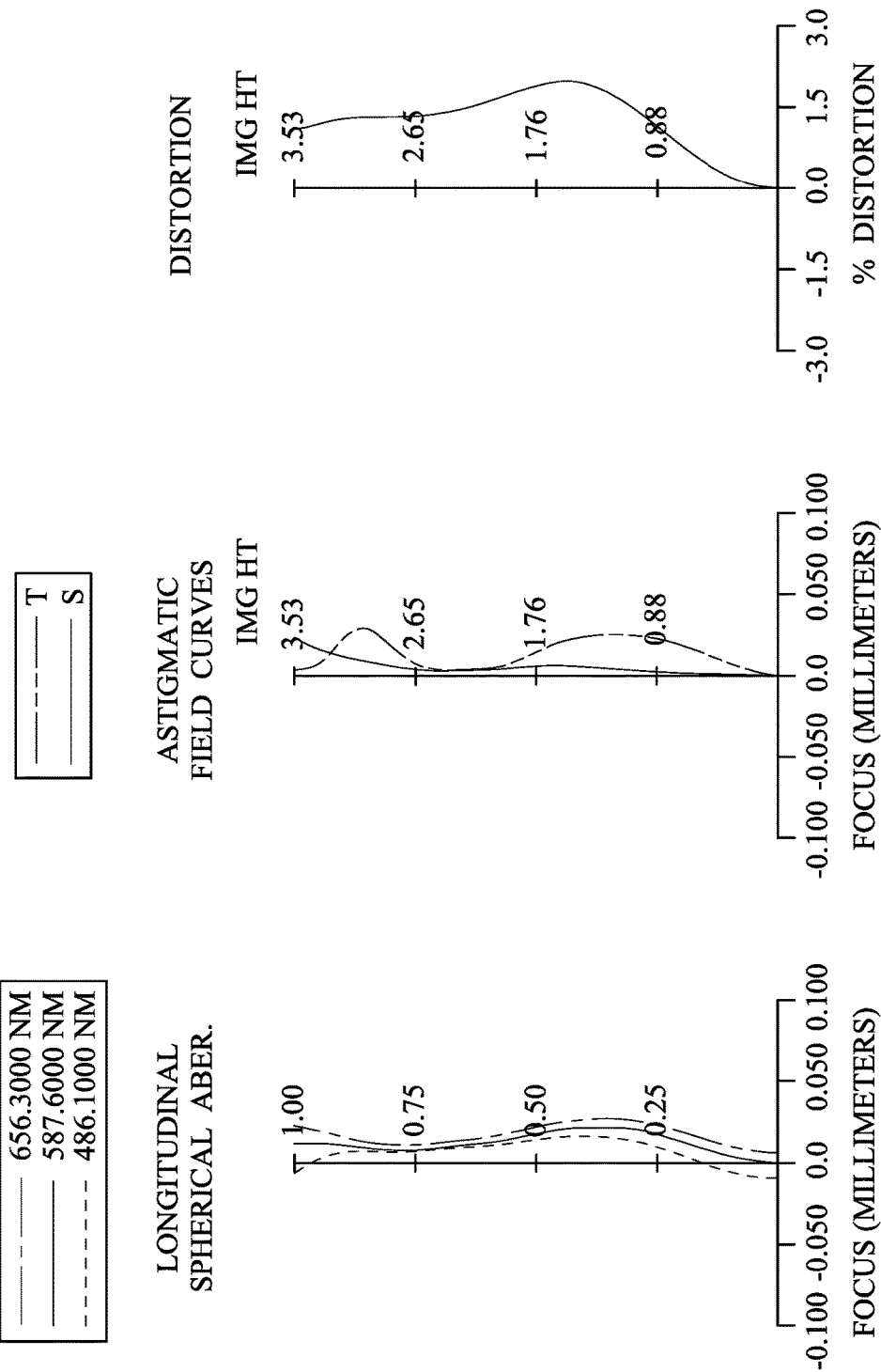


Fig. 2

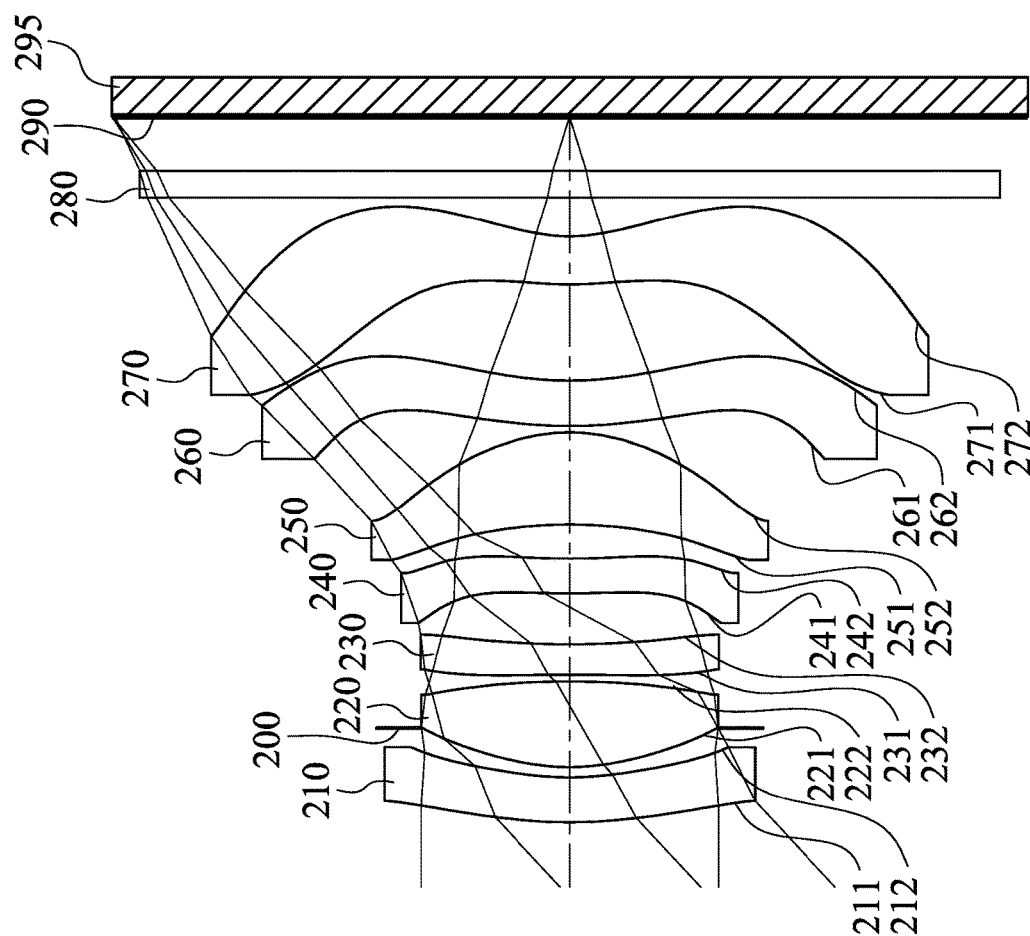


Fig. 3

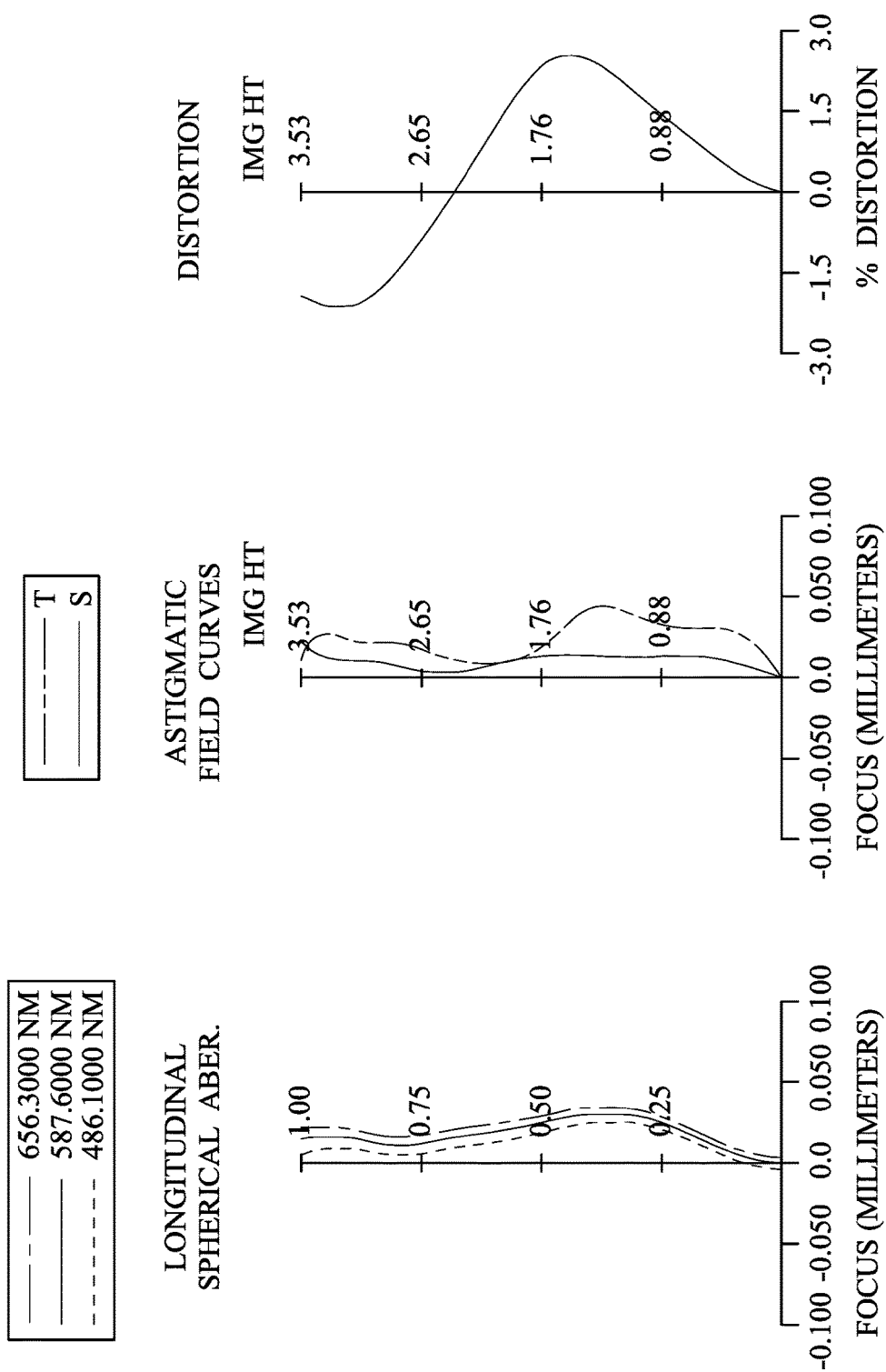


Fig. 4

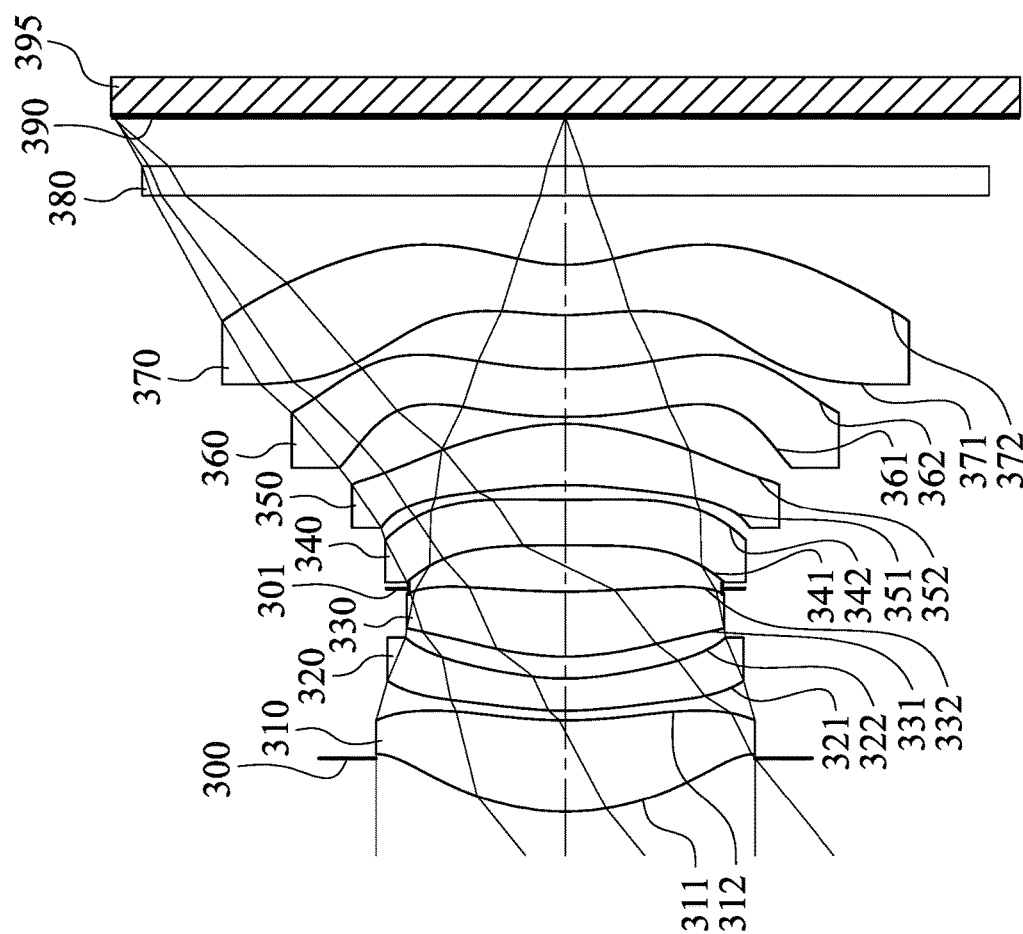


Fig. 5

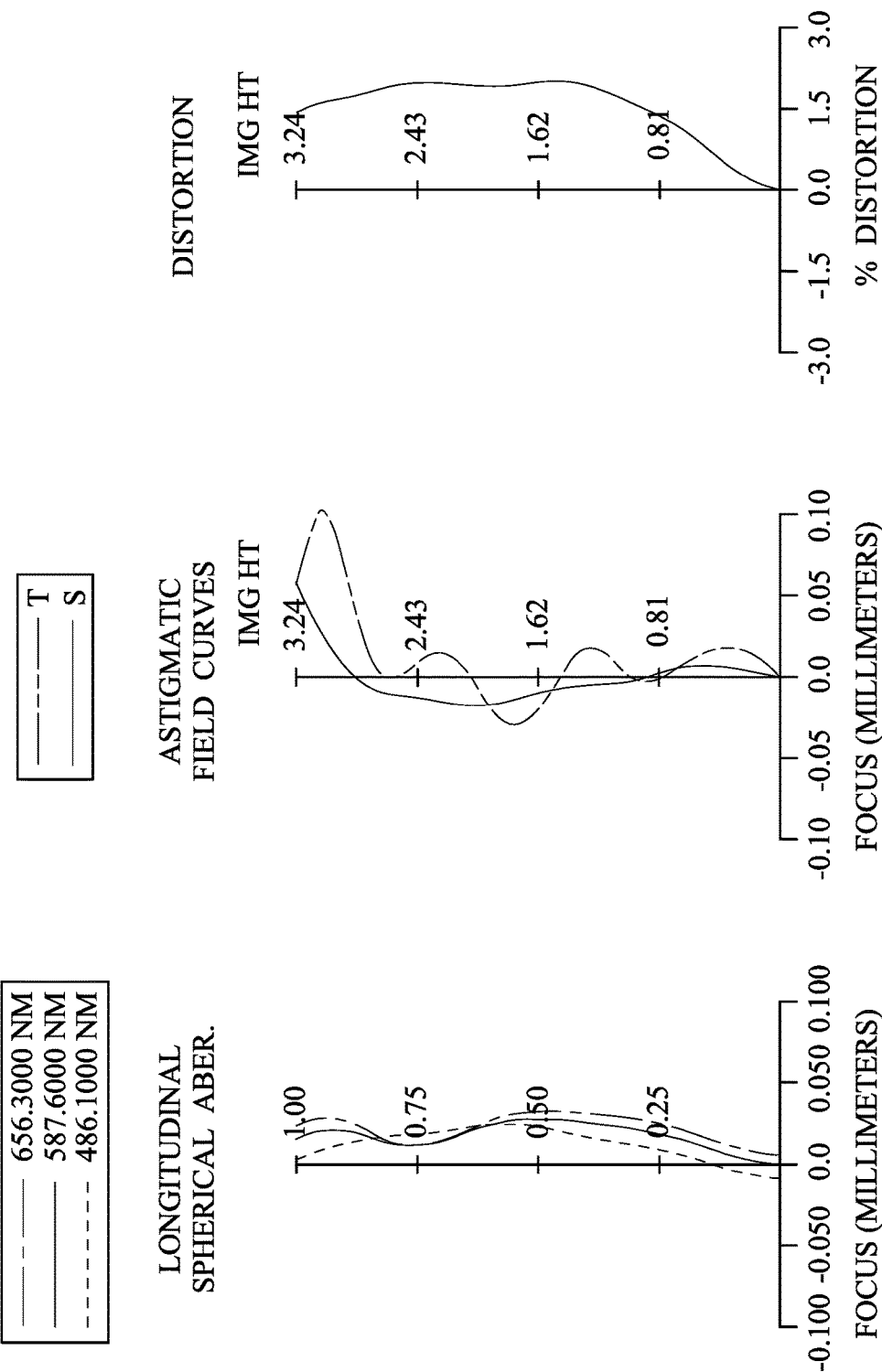


Fig. 6

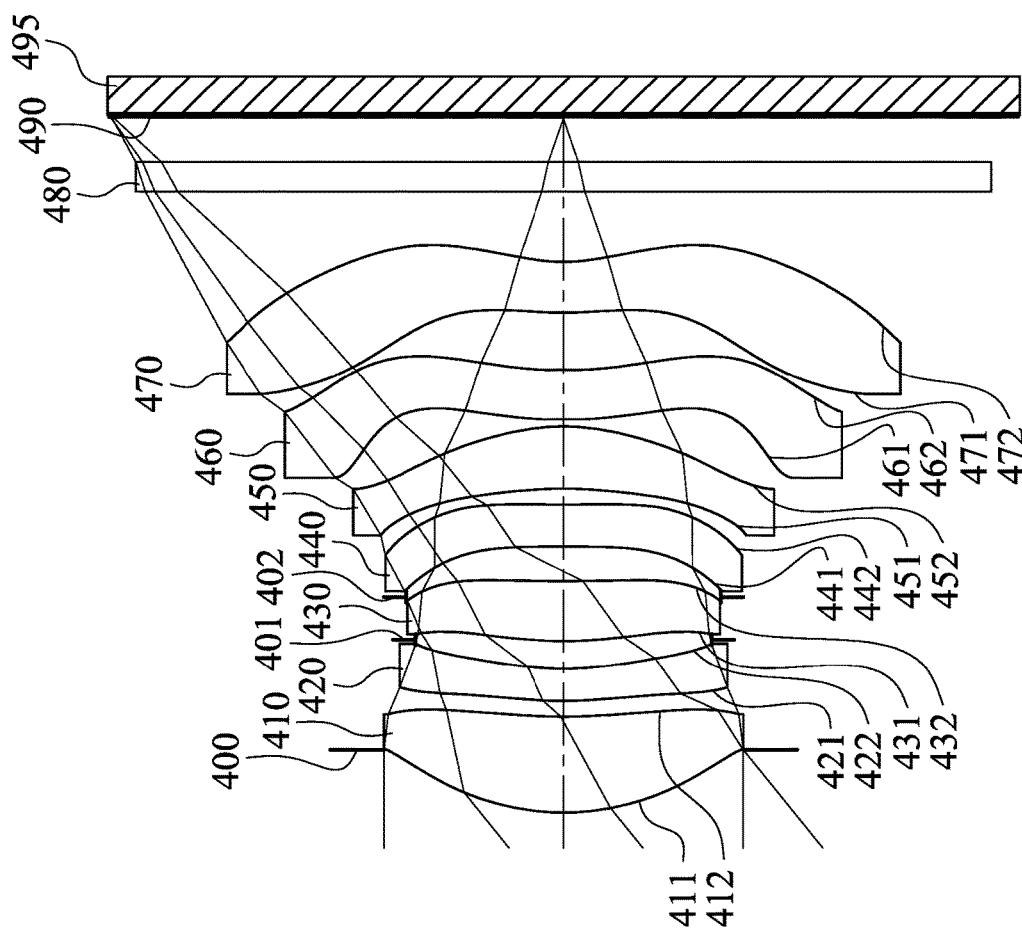
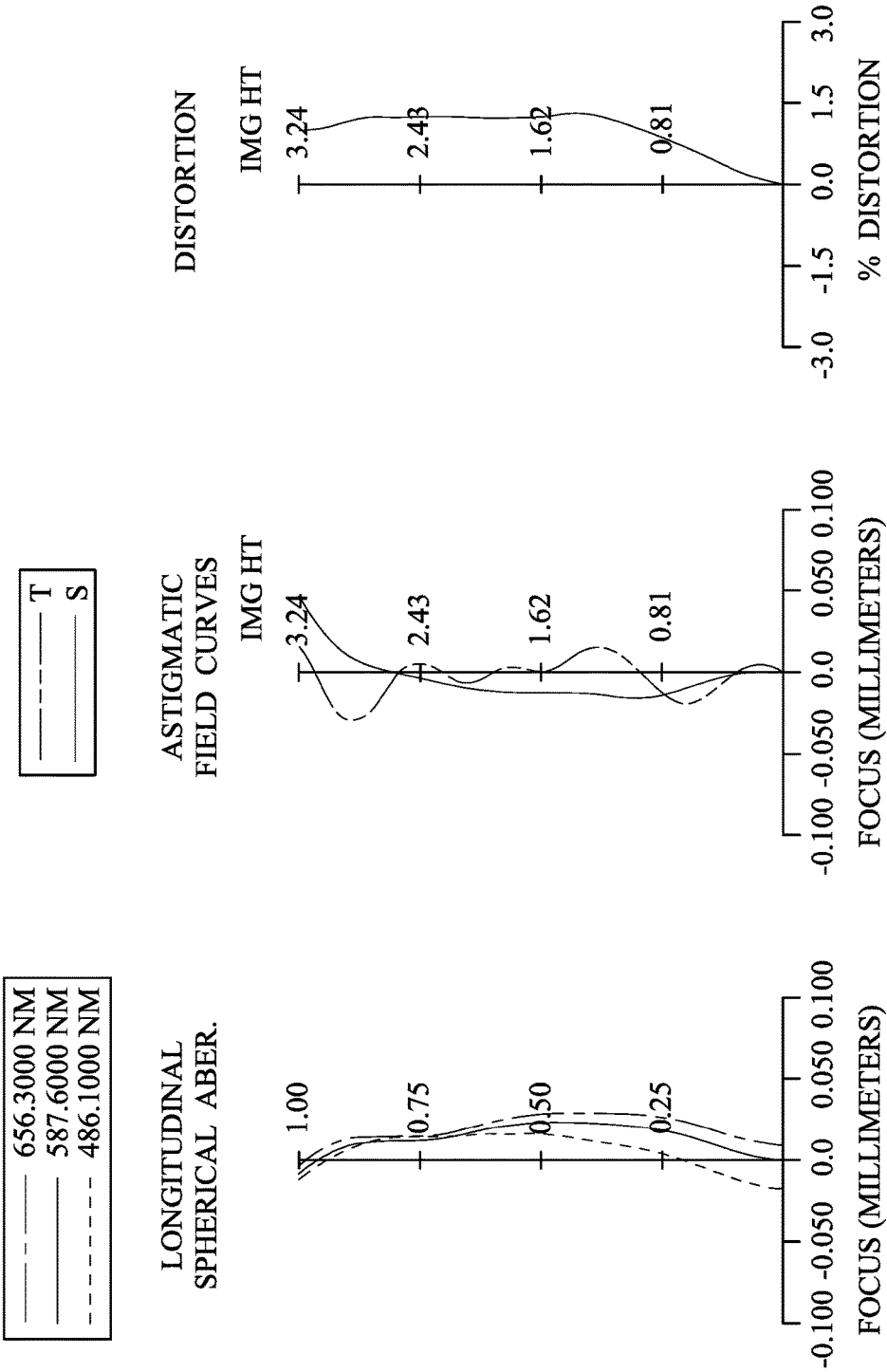


Fig. 7



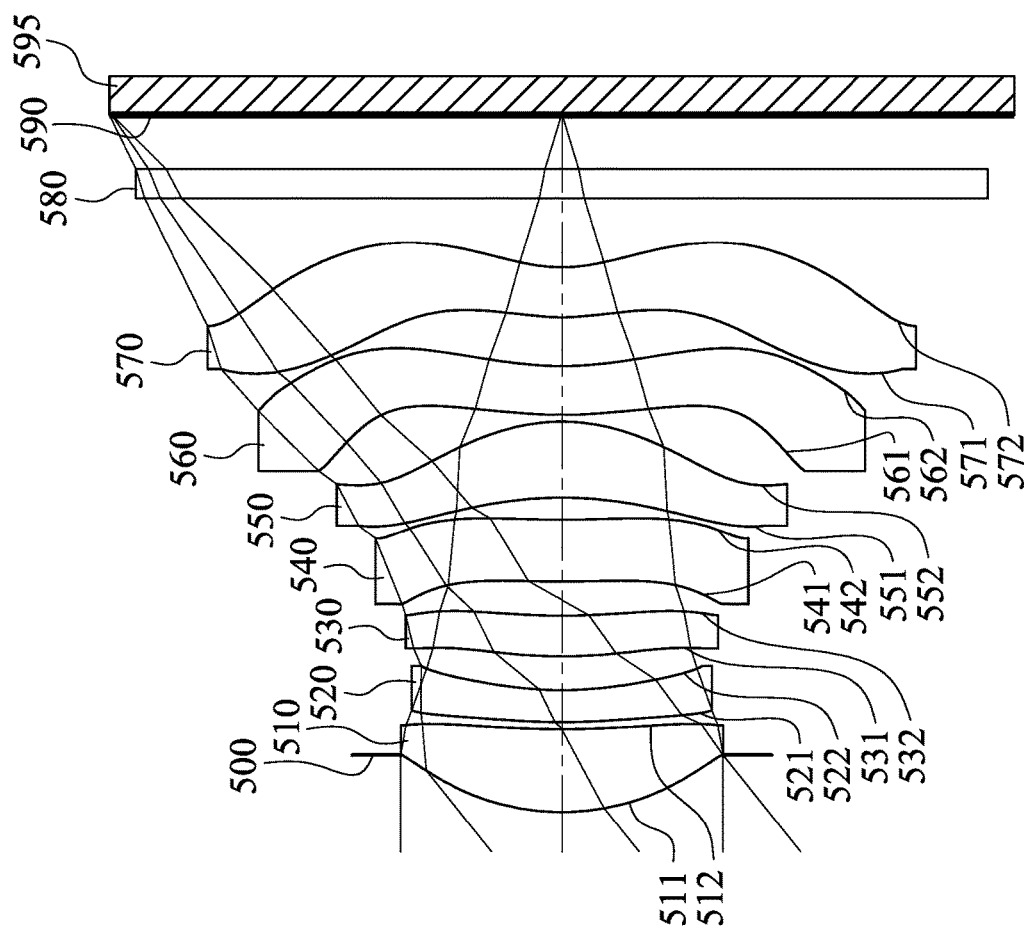


Fig. 9

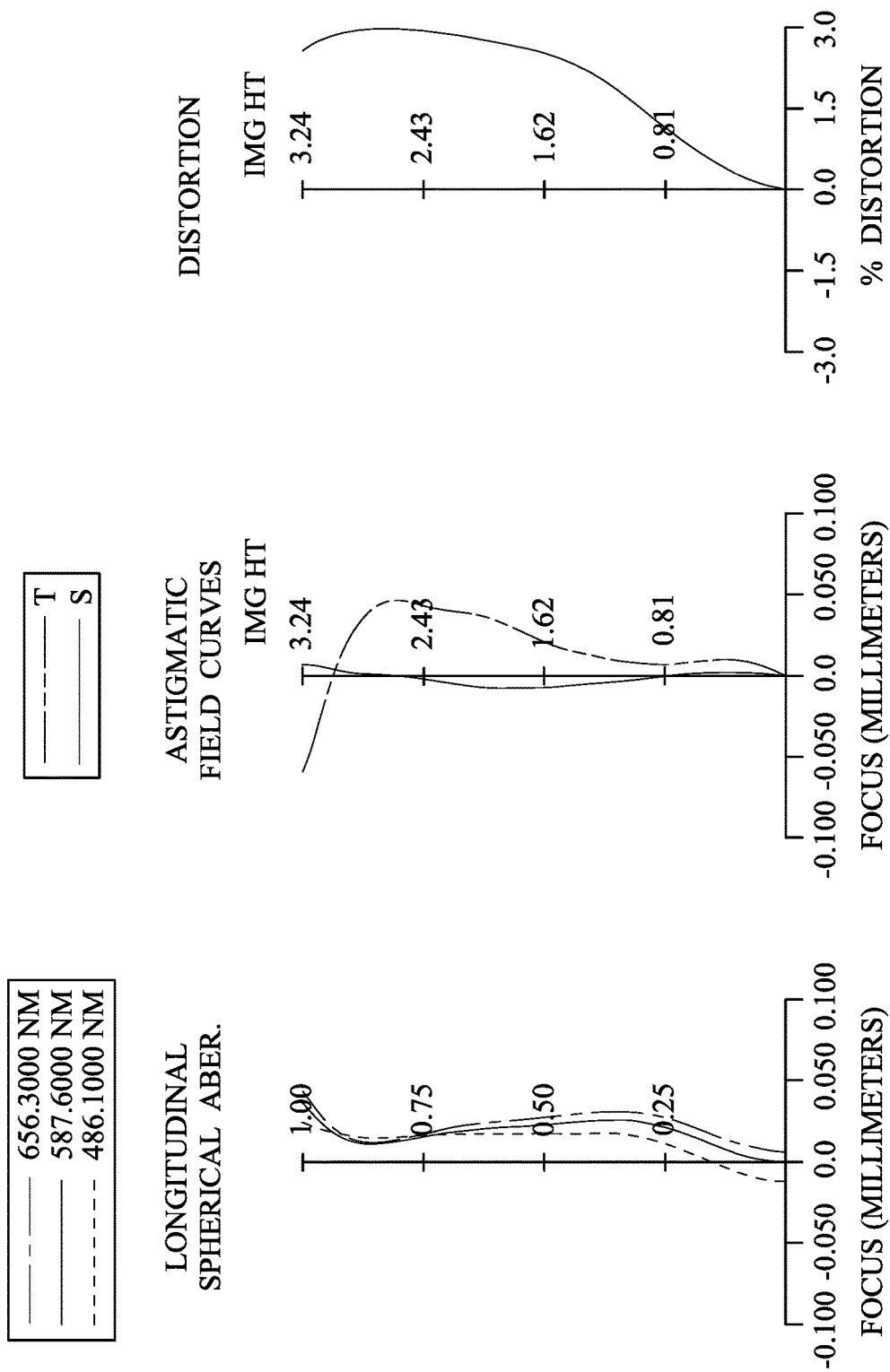


Fig. 10

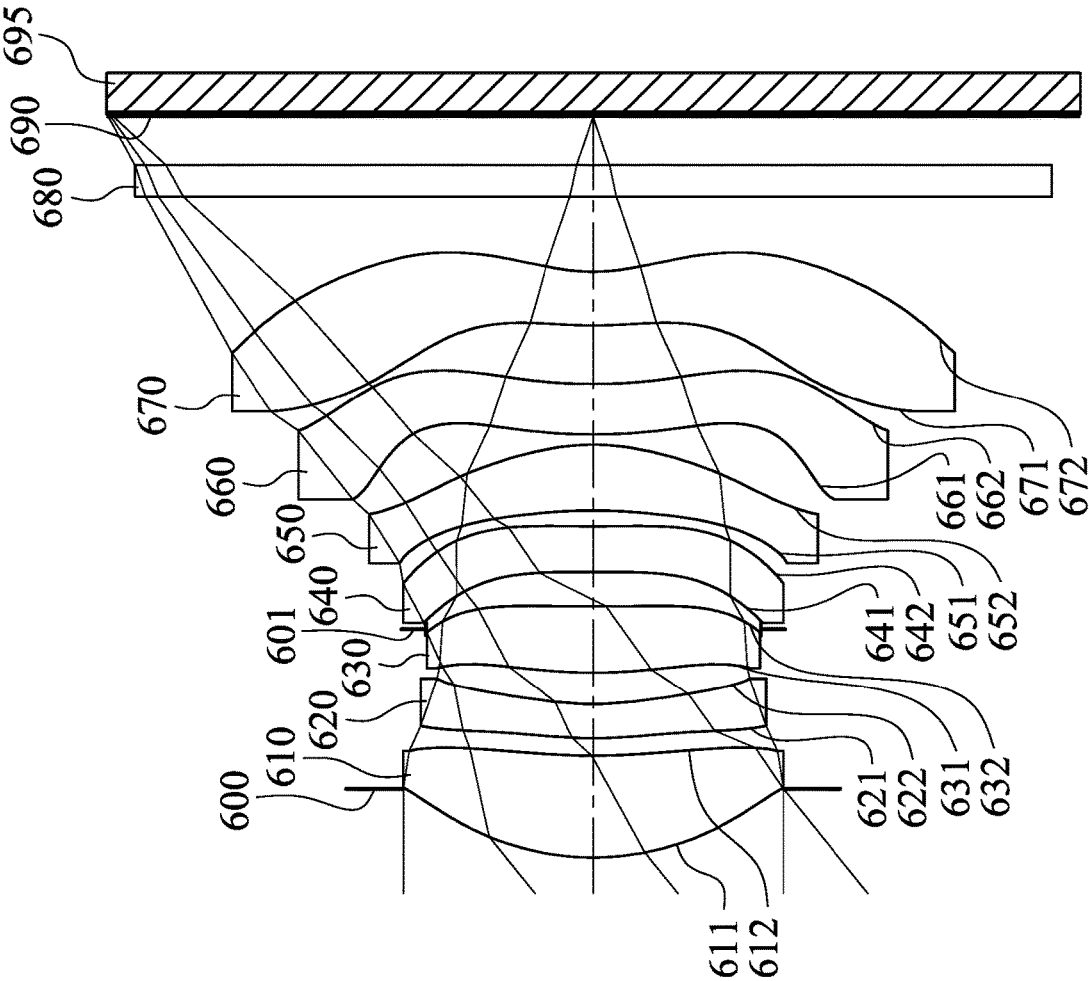


Fig. 11

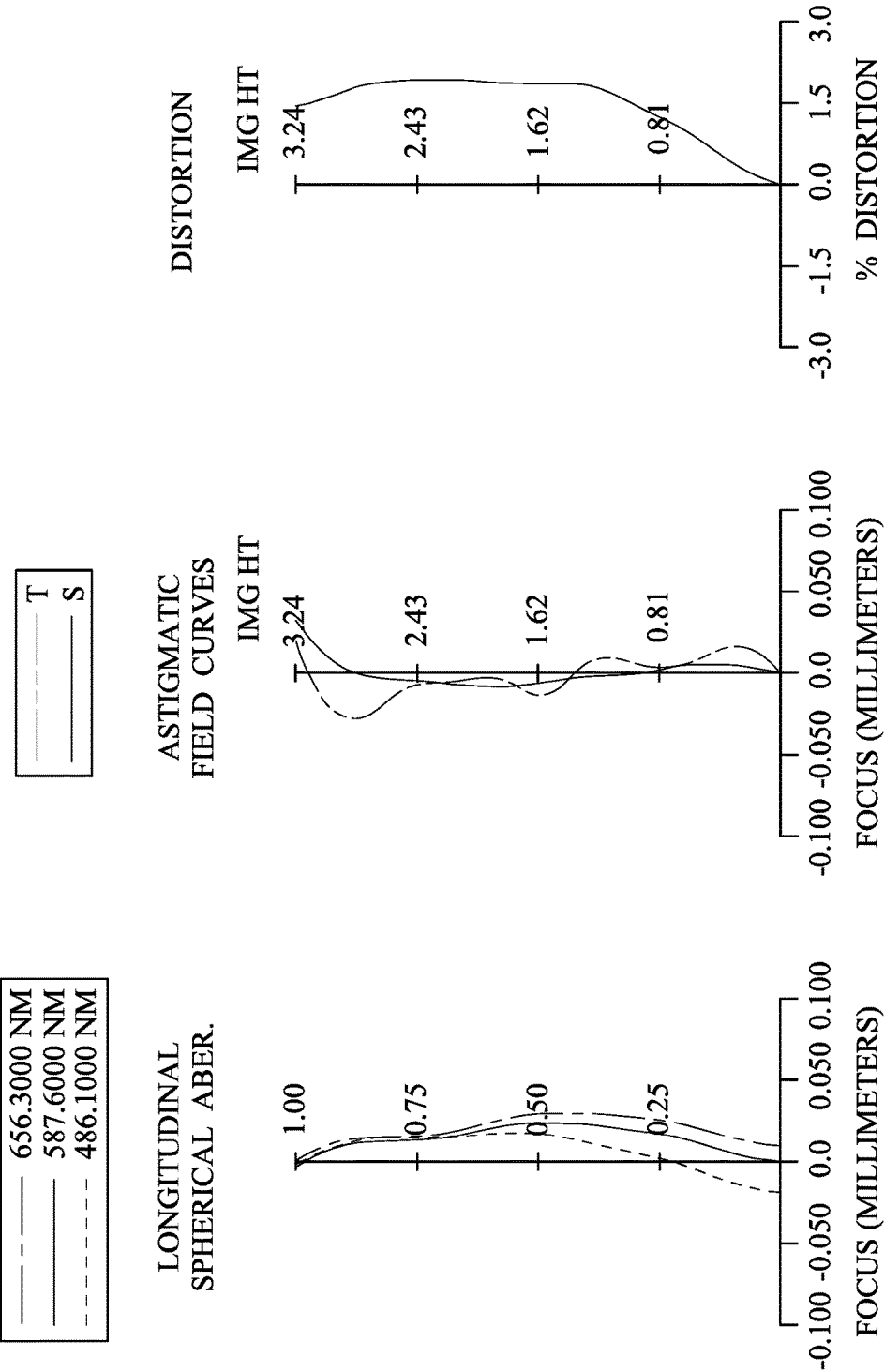


Fig. 12

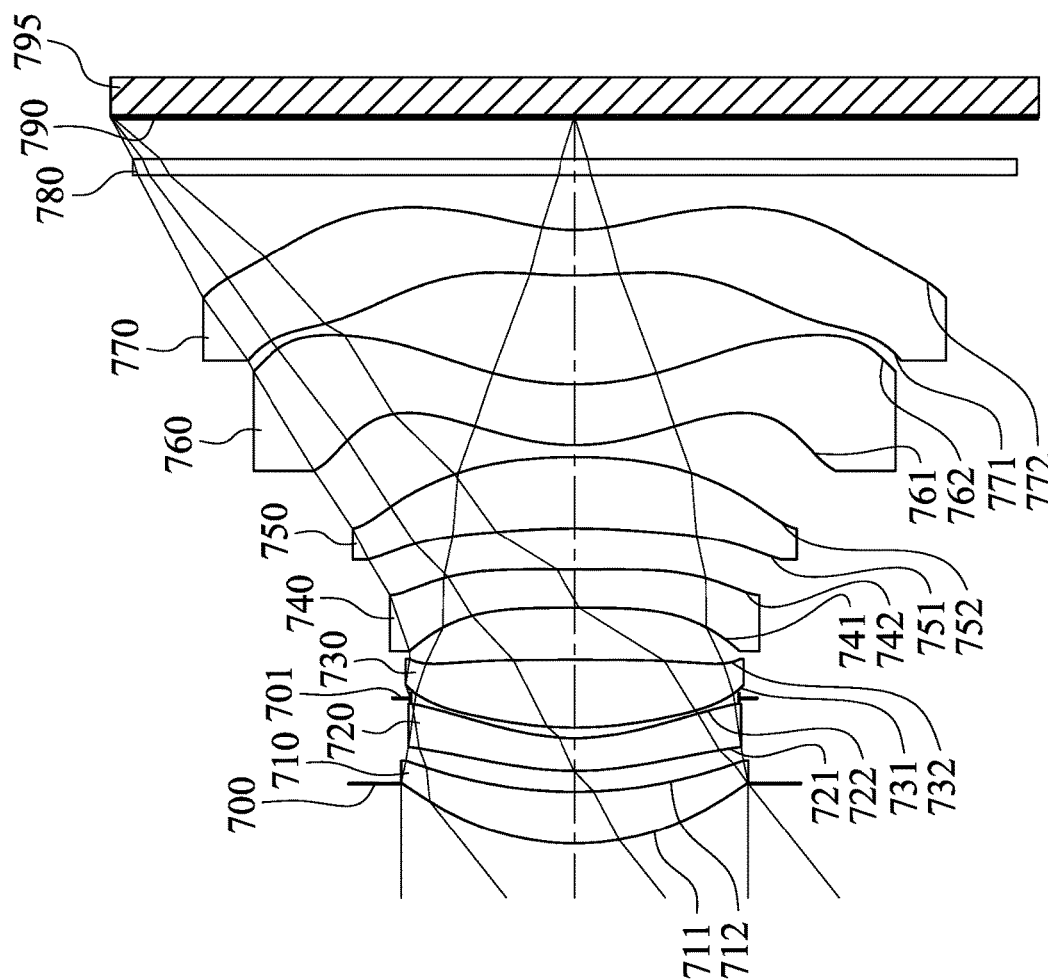


Fig. 13

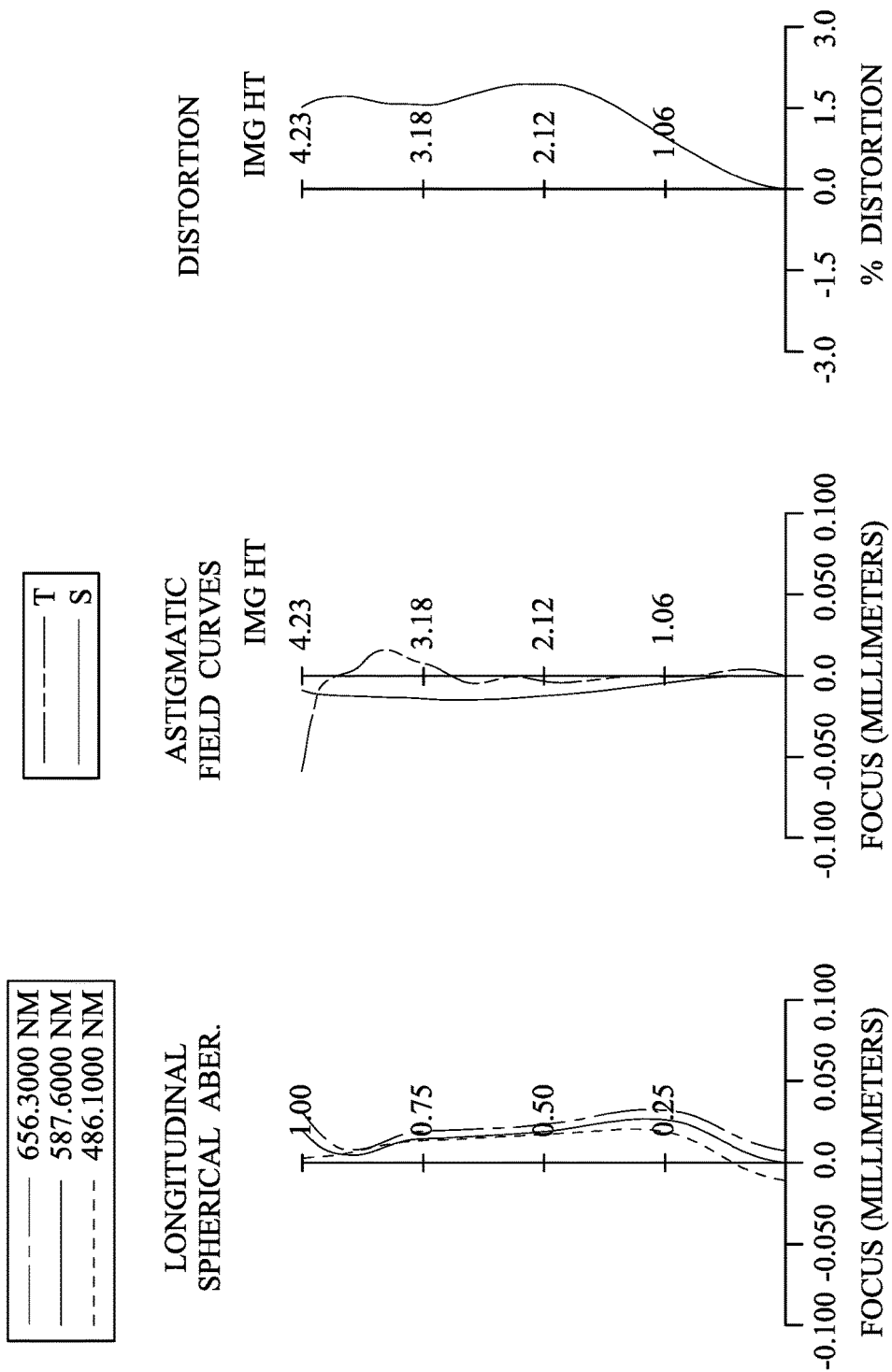
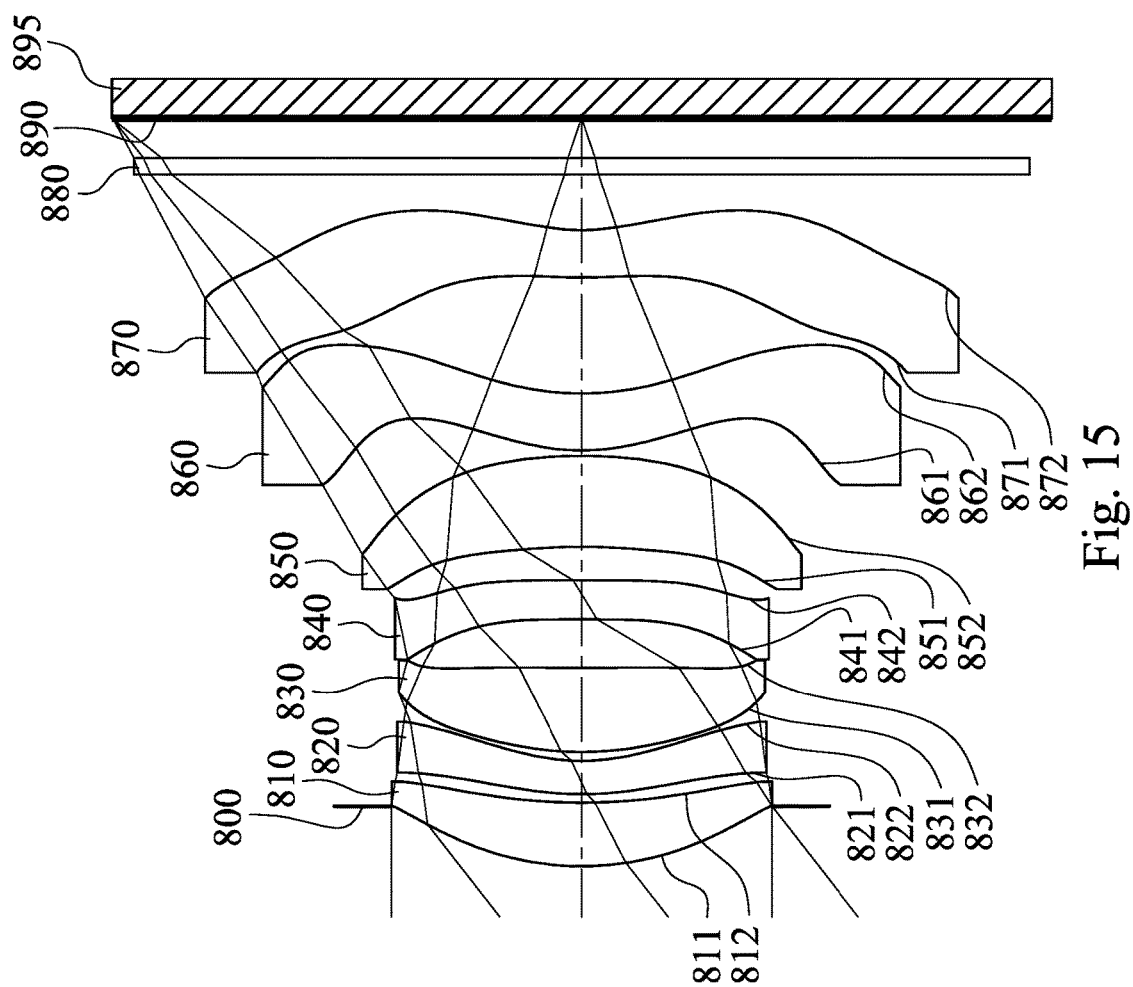


Fig. 14



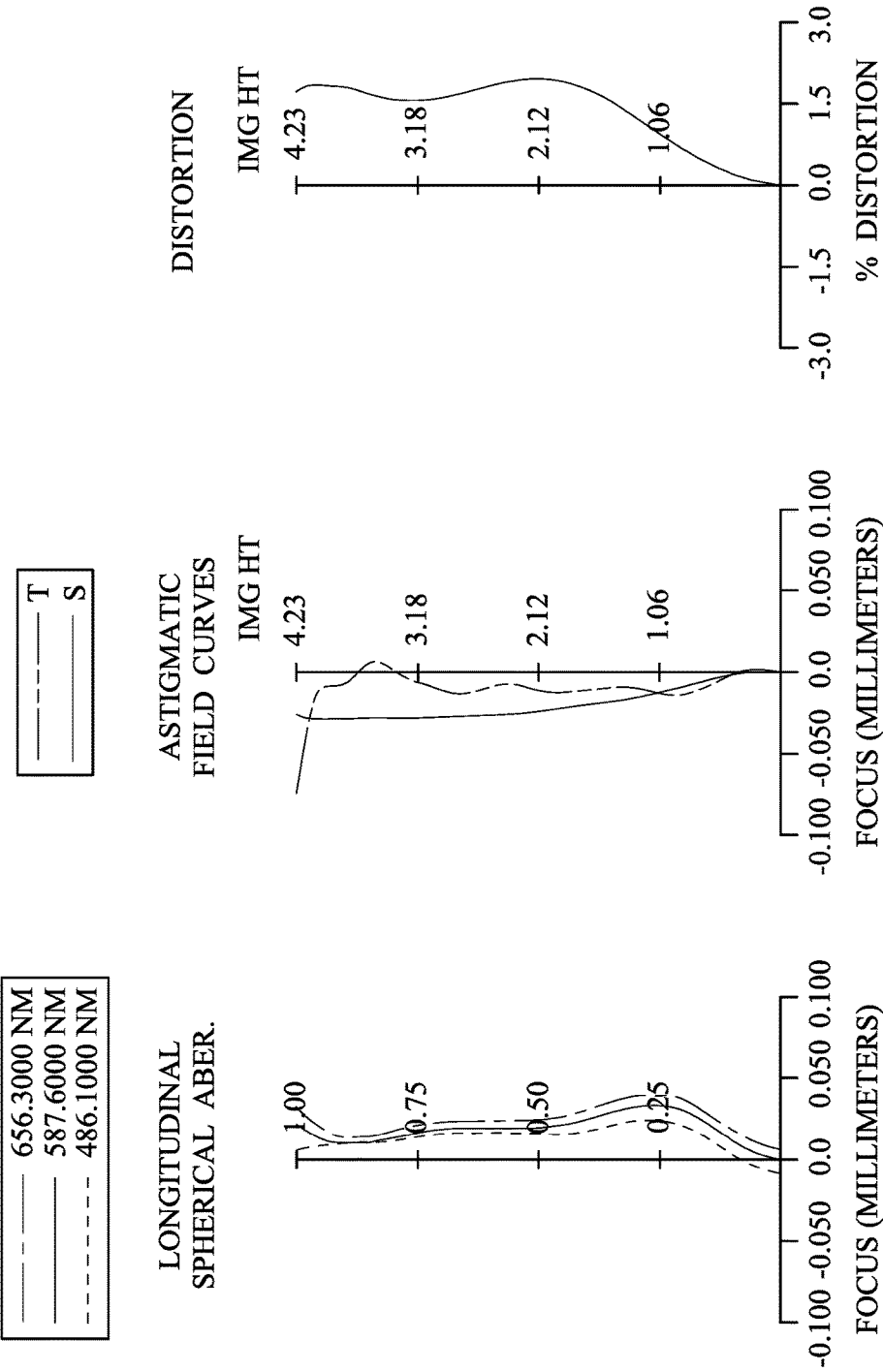


Fig. 16

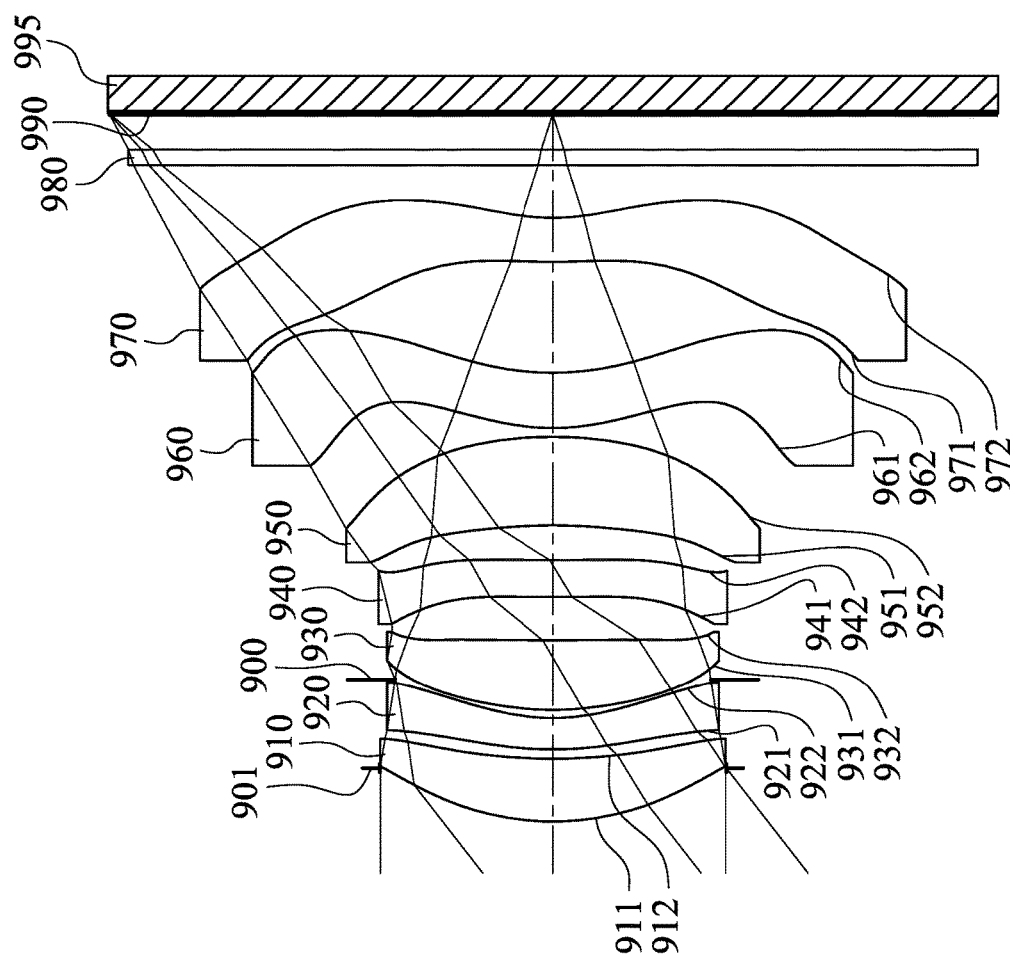


Fig. 17

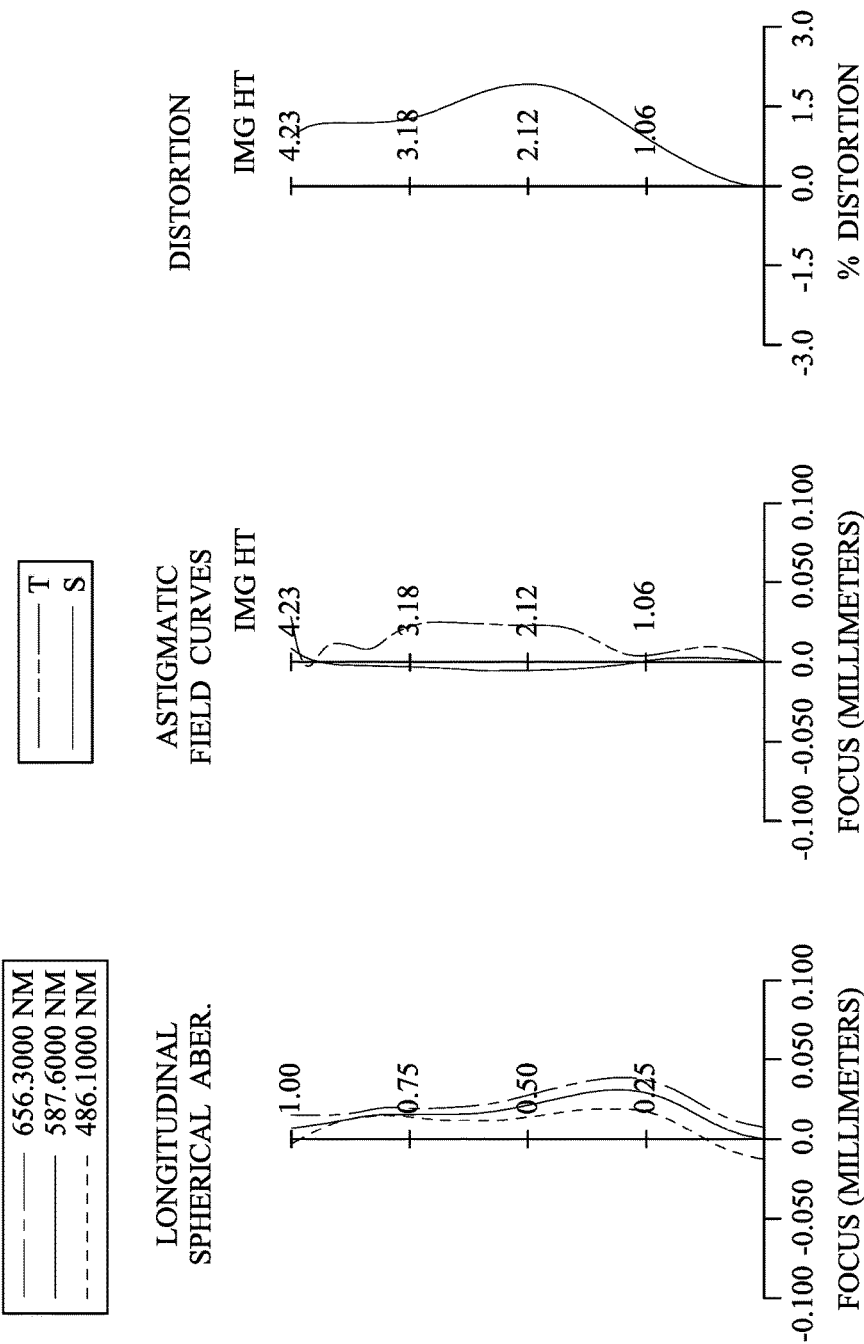


Fig. 18

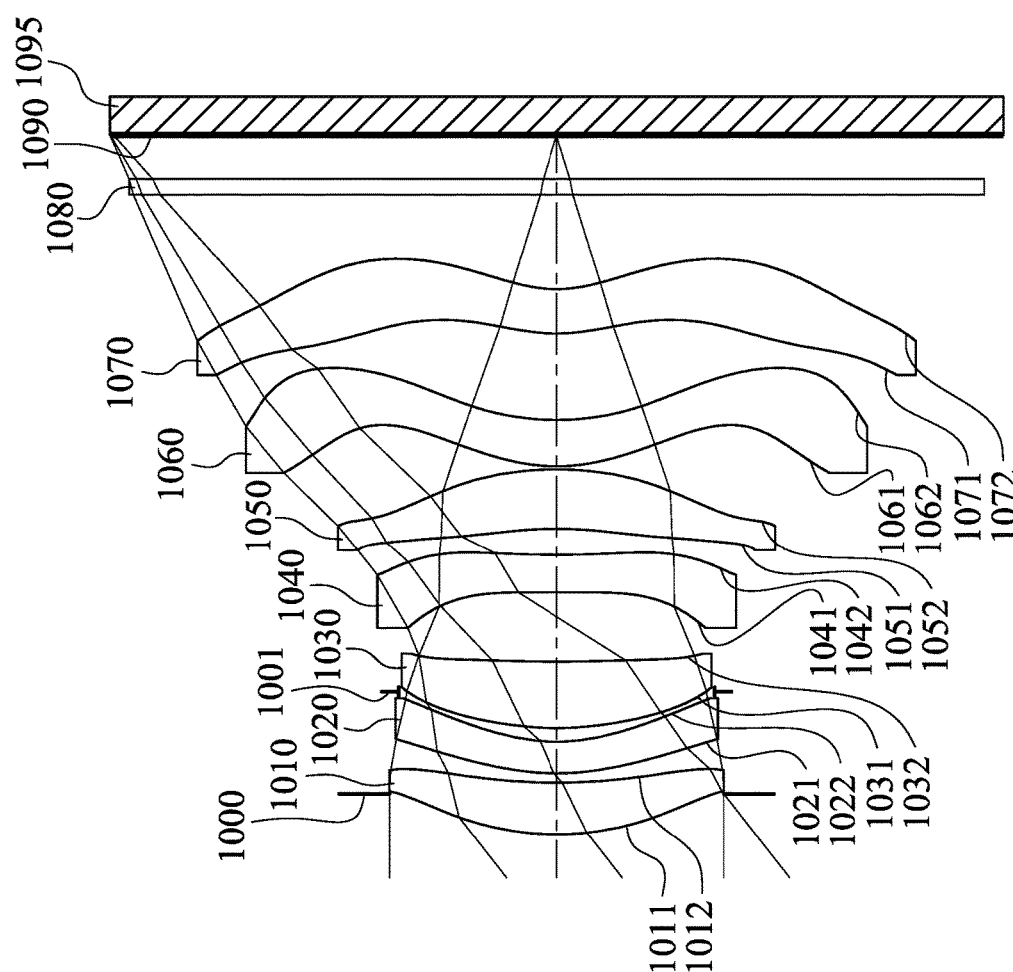


Fig. 19

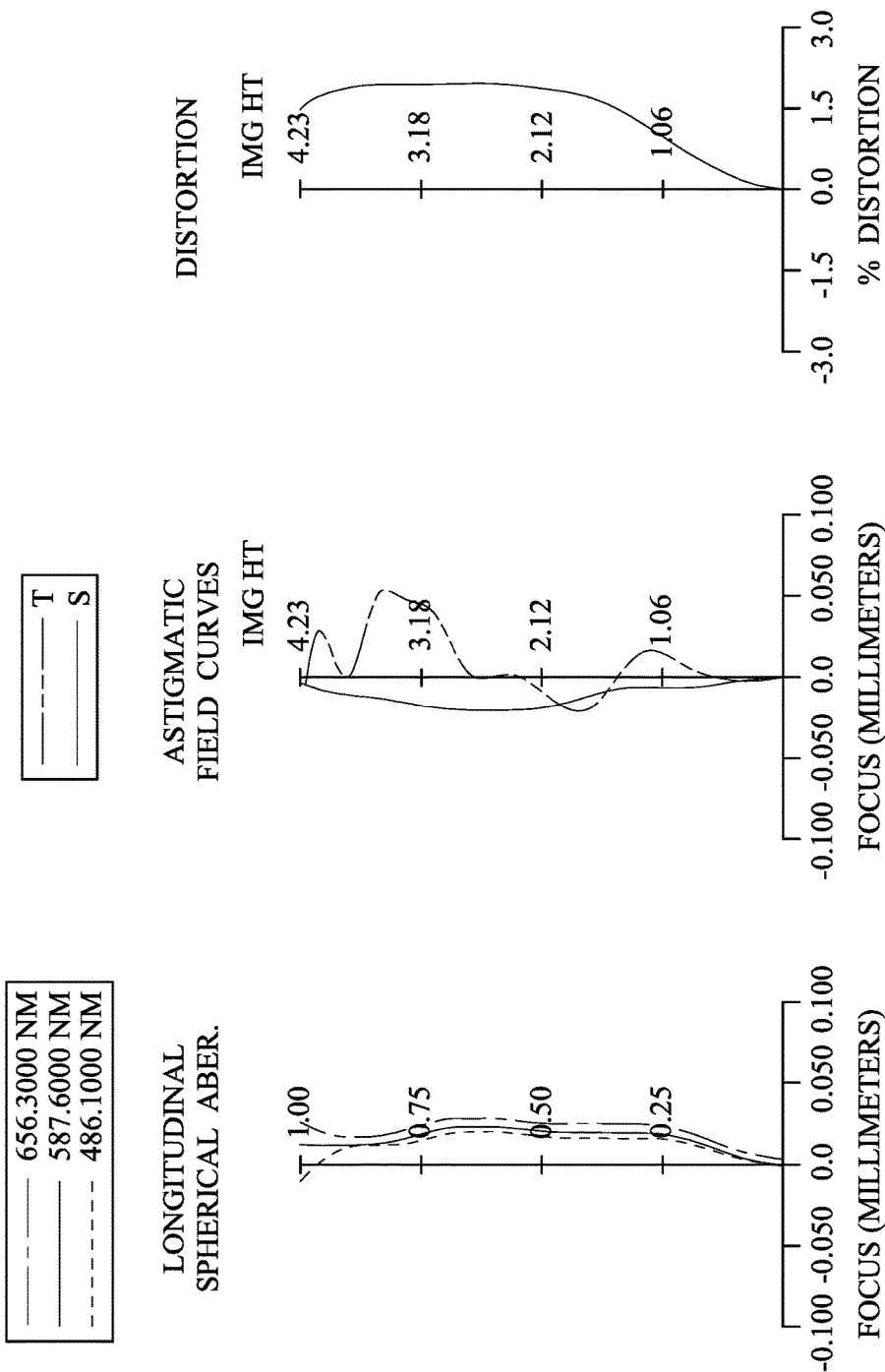


Fig. 20

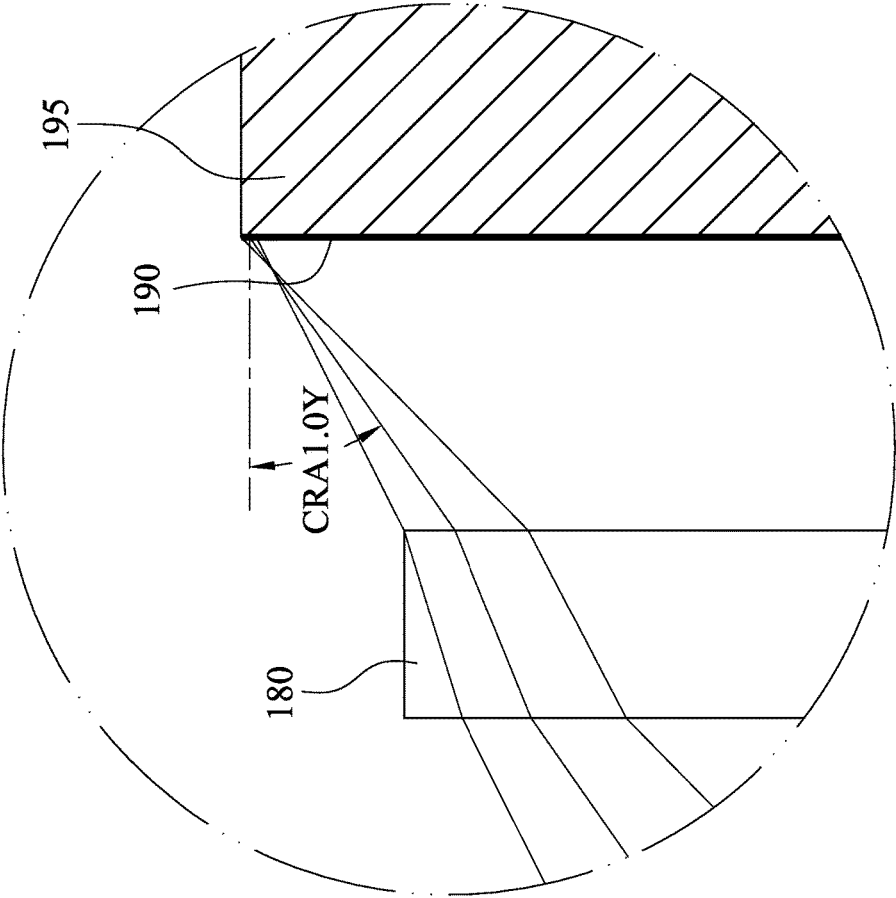


Fig. 21

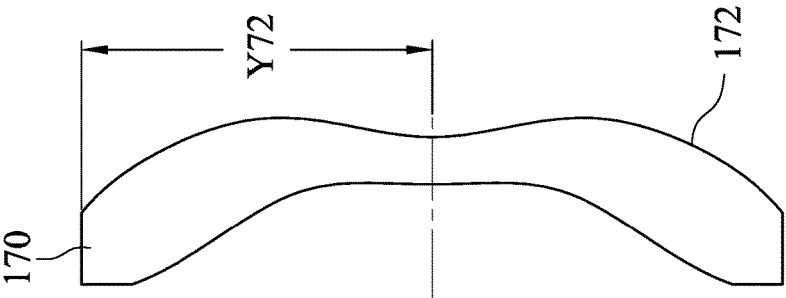


Fig. 22

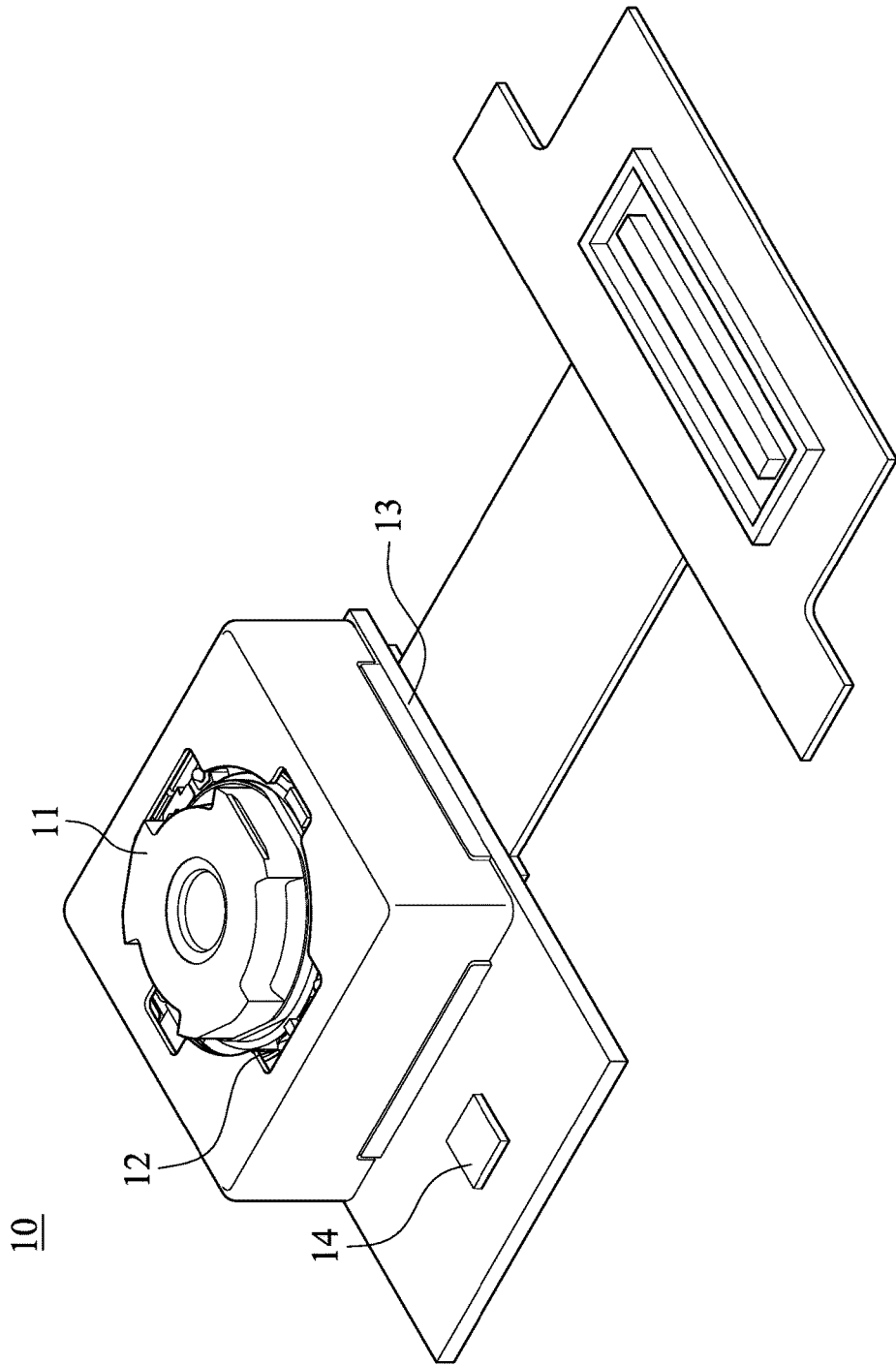


Fig. 23

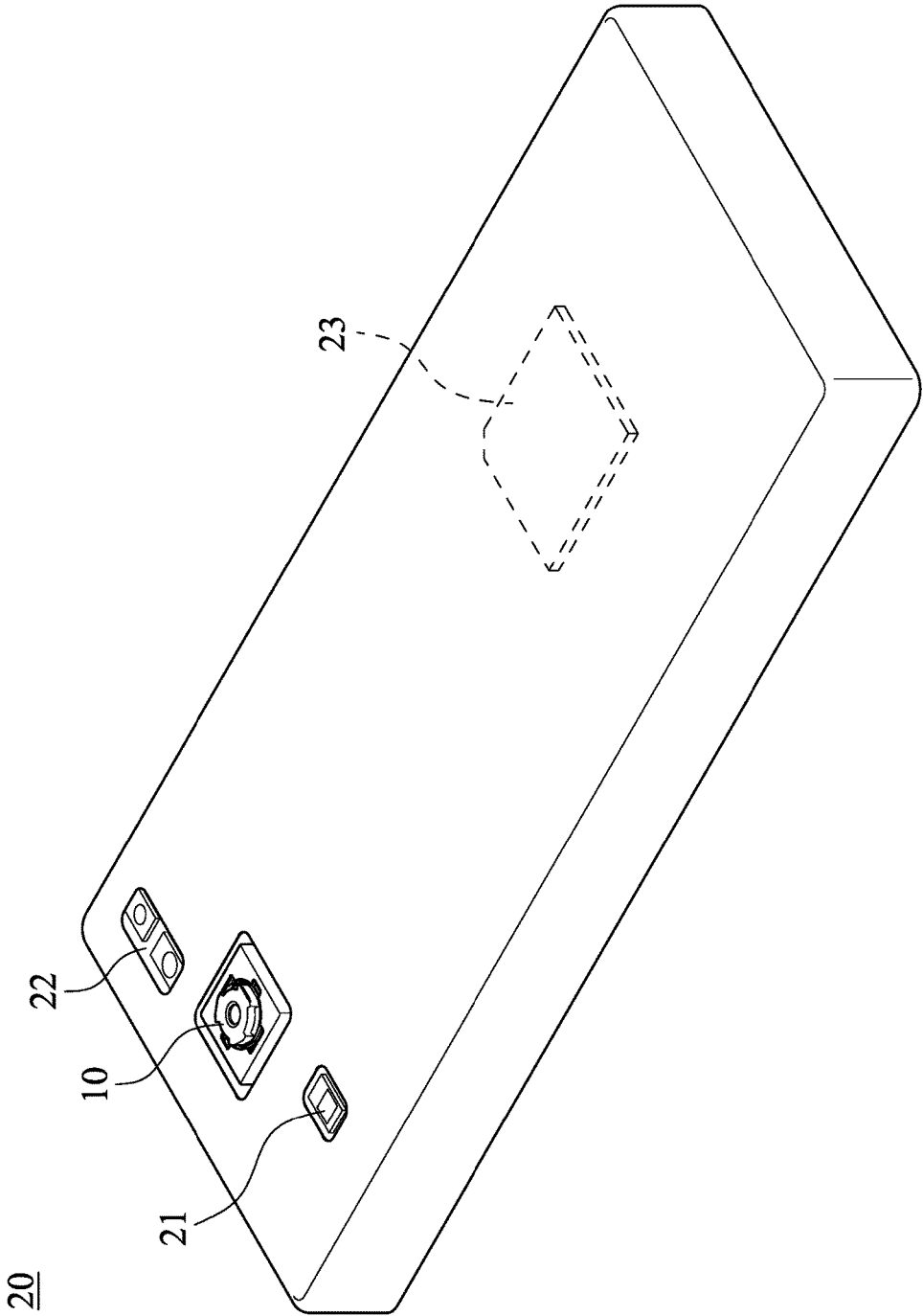


Fig. 24A

20

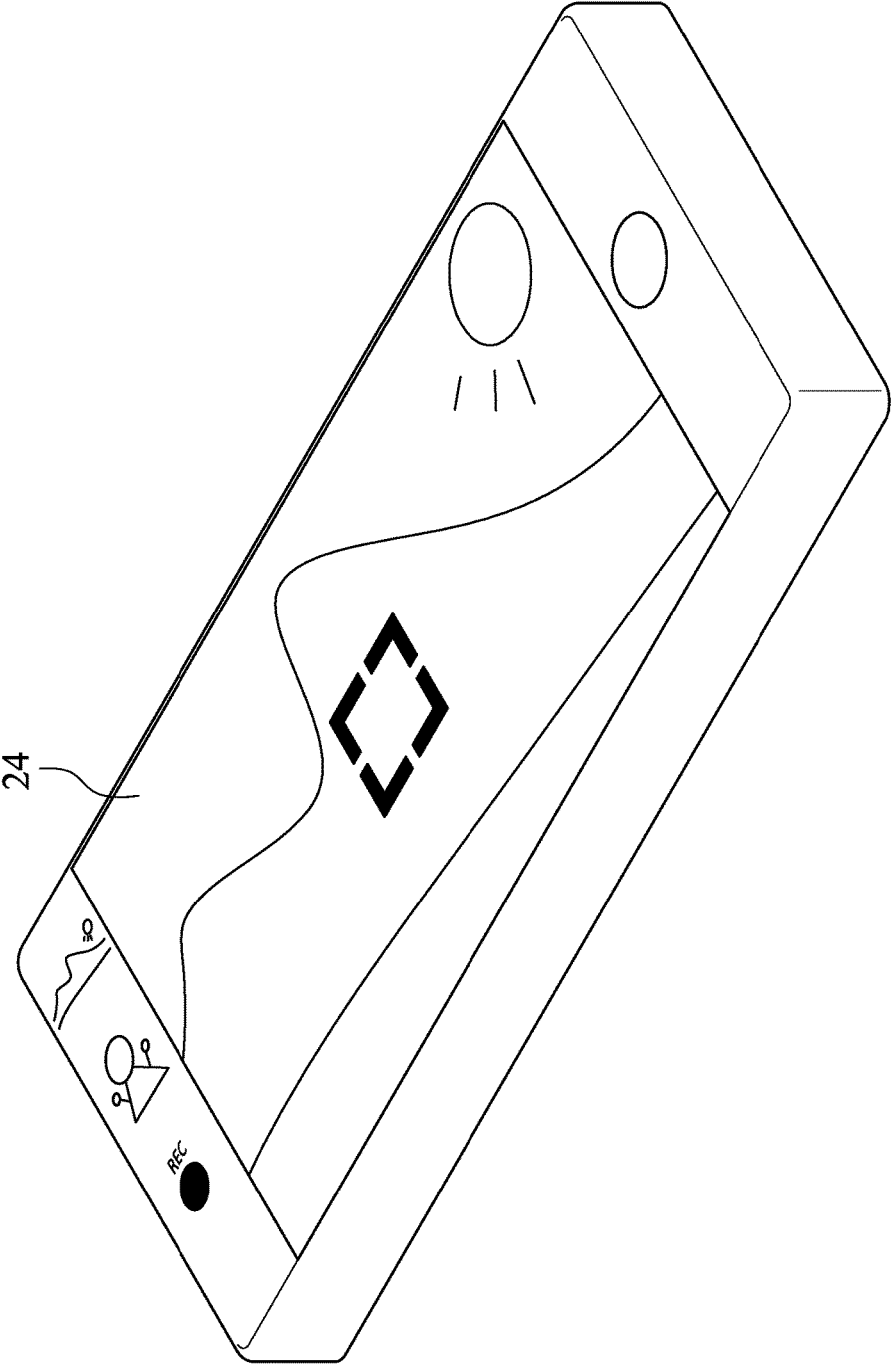


Fig. 24B

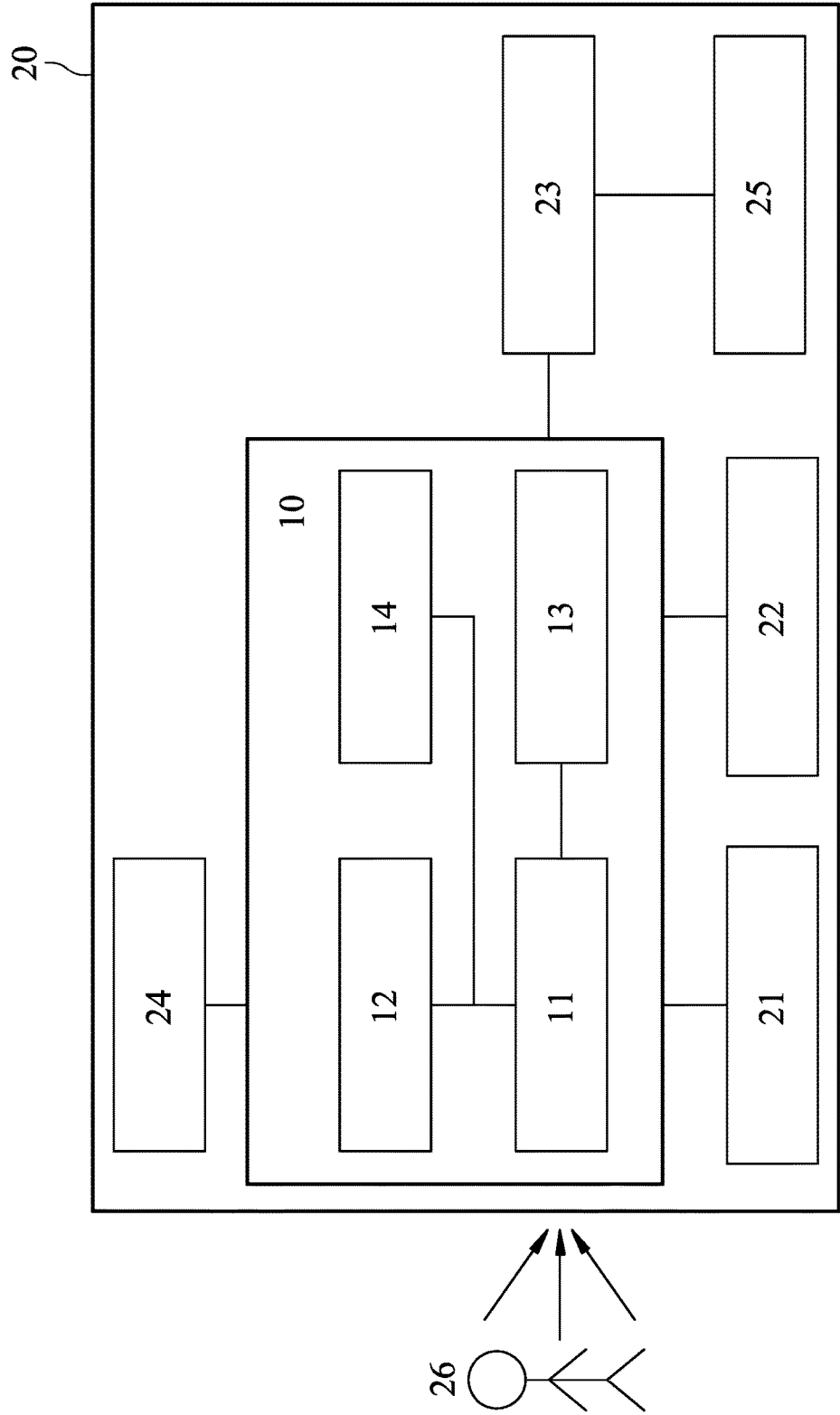


Fig. 24C

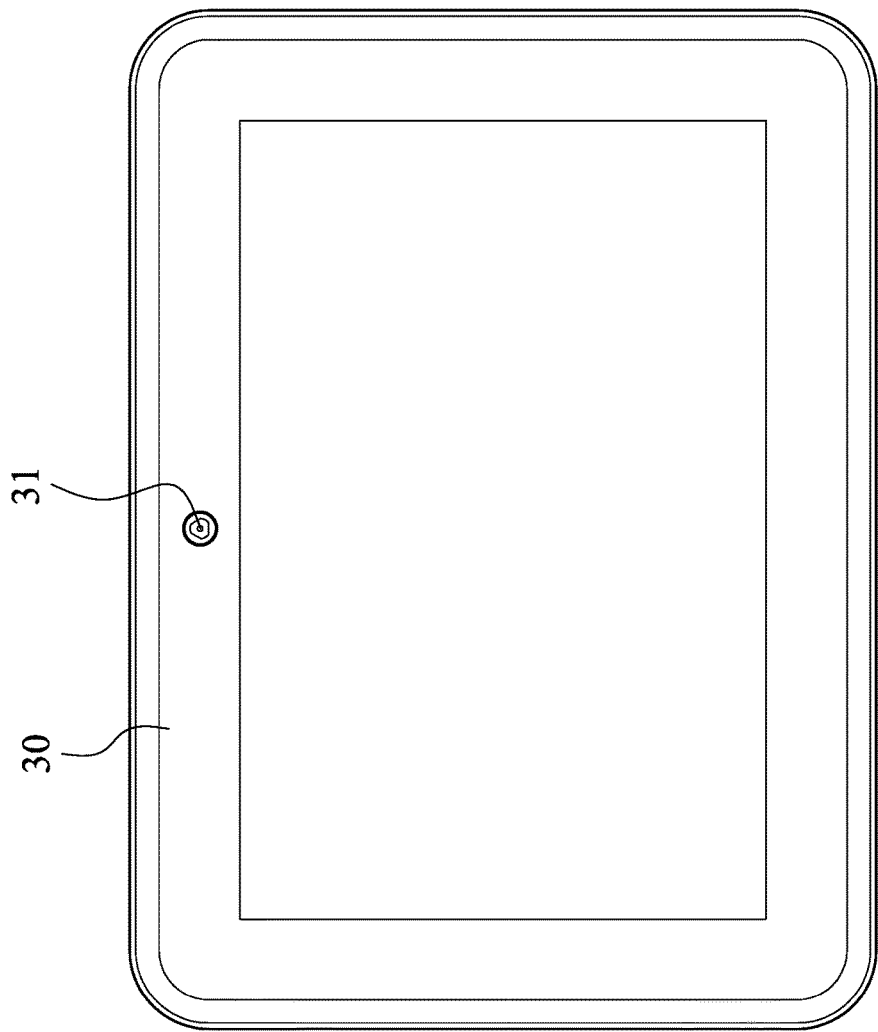


Fig. 25

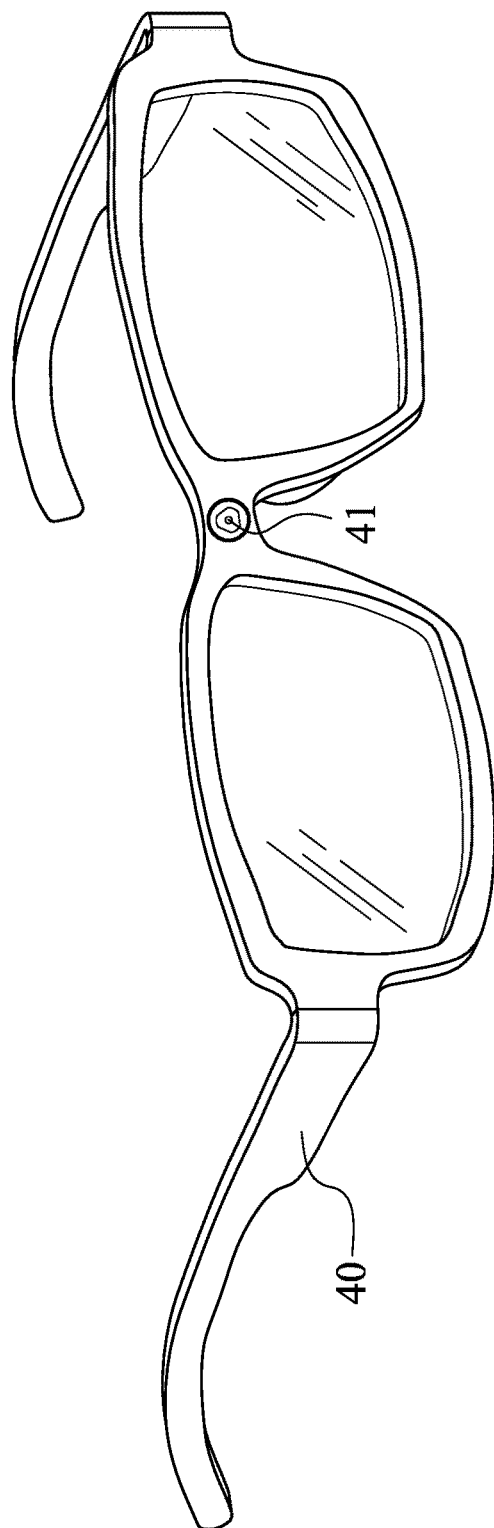


Fig. 26

IMAGE CAPTURING LENS ASSEMBLY, IMAGING APPARATUS AND ELECTRONIC DEVICE

RELATED APPLICATIONS

[0001] This application claims priority to Taiwan Application Serial Number 106118432, filed Jun. 3, 2017, which is herein incorporated by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to an image capturing lens assembly and an imaging apparatus. More particularly, the present disclosure relates to an image capturing lens assembly and an imaging apparatus, each of which is featured with an enhanced imaging functionality in an off-axis region and a short total track length and is applicable to electronic devices.

[0003] Description of Related Art

[0004] With the broadening application of photographing modules, the installation of the photographing modules in various intelligent electronic products, entertainment devices, fitness devices and smart home systems is an upward trend of future technical development. However, with the advance of science and technology, as well as the improved functionality of the electronic products, such as smartphone, the demand of consumers for camera functions is also increasing (such as night photography, photographs of fast motion or depth of focus). It is known that conventional optical lens modules have difficulty satisfying the demands of a large aperture and short total track length simultaneously. Particularly, the result of photographing the off-axis region of an image is highly dictated by the aperture size. Therefore, enhancing the imaging functionality in the off-axis region while maintaining a short total track length, which enables the photographing modules to be applicable to portable electronic devices, is the goal for manufacturers in the industry.

SUMMARY

[0005] According to one aspect of the present disclosure, an image capturing lens assembly includes seven lens elements, which are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element. The fifth lens element with positive refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. The sixth lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, and the object-side surface and the image-side surface of the sixth lens element are aspheric. The seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, the image-side surface of the seventh lens element includes at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the seventh lens element are aspheric. When an axial distance between an object-side surface of the first lens element and an image surface is TL, and an entrance pupil

diameter of the image capturing lens assembly is EPD, the following condition is satisfied:

$$1.0 < TL/EPD \leq 2.40.$$

[0006] According to another aspect of the present disclosure, an imaging apparatus includes the image capturing lens assembly according to the aforementioned aspect and an image sensor, wherein the image sensor is disposed on the image surface of the image capturing lens assembly.

[0007] According to further another aspect of the present disclosure, an electronic device includes the imaging apparatus according to the aforementioned aspect.

[0008] According to yet another aspect of the present disclosure, an image capturing lens assembly includes seven lens elements, which are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element. The fifth lens element has positive refractive power. The sixth lens element has an image-side surface being concave in a paraxial region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, the image-side surface of the seventh lens element includes at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the seventh lens element are aspheric. When an axial distance between an object-side surface of the first lens element and an image surface is TL, and an entrance pupil diameter of the image capturing lens assembly is EPD, the following condition is satisfied:

$$1.20 < TL/EPD \leq 2.10.$$

[0009] According to still another aspect of the present disclosure, an image capturing lens assembly includes seven lens elements, which are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element. The sixth lens element has an image-side surface being concave in a paraxial region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, the image-side surface of the seventh lens element includes at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the seventh lens element are aspheric. When an axial distance between an object-side surface of the first lens element and an image surface is TL, an entrance pupil diameter of the image capturing lens assembly is EPD, an axial distance between the second lens element and the third lens element is T23, and an axial distance between the third lens element and the fourth lens element is T34, the following conditions are satisfied:

$$1.0 < TL/EPD \leq 2.10; \text{ and}$$

$$0 < T23/T34 \leq 0.90.$$

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0011] FIG. 1 is a schematic view of an imaging apparatus according to the 1st embodiment of the present disclosure;

[0012] FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 1st embodiment;

[0013] FIG. 3 is a schematic view of an imaging apparatus according to the 2nd embodiment of the present disclosure;

[0014] FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 2nd embodiment;

[0015] FIG. 5 is a schematic view of an imaging apparatus according to the 3rd embodiment of the present disclosure;

[0016] FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 3rd embodiment;

[0017] FIG. 7 is a schematic view of an imaging apparatus according to the 4th embodiment of the present disclosure;

[0018] FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 4th embodiment;

[0019] FIG. 9 is a schematic view of an imaging apparatus according to the 5th embodiment of the present disclosure;

[0020] FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 5th embodiment;

[0021] FIG. 11 is a schematic view of an imaging apparatus according to the 6th embodiment of the present disclosure;

[0022] FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 6th embodiment;

[0023] FIG. 13 is a schematic view of an imaging apparatus according to the 7th embodiment of the present disclosure;

[0024] FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 7th embodiment;

[0025] FIG. 15 is a schematic view of an imaging apparatus according to the 8th embodiment of the present disclosure;

[0026] FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 8th embodiment;

[0027] FIG. 17 is a schematic view of an imaging apparatus according to the 9th embodiment of the present disclosure;

[0028] FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 9th embodiment;

[0029] FIG. 19 is a schematic view of an imaging apparatus according to the 10th embodiment of the present disclosure;

[0030] FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 10th embodiment;

[0031] FIG. 21 is a schematic view showing a parameter of CRA1.0Y of the imaging apparatus according to the 1st embodiment in FIG. 1;

[0032] FIG. 22 is a schematic view showing a parameter of Y72 of the imaging apparatus according to the 1st embodiment in FIG. 1;

[0033] FIG. 23 is a three-dimensional view of an imaging apparatus according to the 11th embodiment of the present disclosure;

[0034] FIG. 24A is a schematic view showing a side of an electronic device according to the 12th embodiment of the present disclosure;

[0035] FIG. 24B is a schematic view showing another side of the electronic device in FIG. 24A;

[0036] FIG. 24C is a block diagram of the electronic device in FIG. 24A;

[0037] FIG. 25 is a schematic view of an electronic device according to the 13th embodiment of the present disclosure; and

[0038] FIG. 26 is a schematic view of an electronic device according to the 14th embodiment of the present disclosure.

DETAILED DESCRIPTION

[0039] An image capturing lens assembly includes seven lens elements, which are, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element.

[0040] The first lens element can have positive refractive power, and can have an object-side surface being convex in a paraxial region thereof. Therefore, the light converging ability at the object side of the image capturing lens assembly can be provided, which is favorable for shortening the total track length of the image capturing lens assembly so as to obtain a compact size.

[0041] The fourth lens element can have an object-side surface being concave in a paraxial region thereof. Therefore, it is favorable for correcting aberrations of the image capturing lens assembly.

[0042] The fifth lens element can have positive refractive power, and can have an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. Therefore, the distribution of the positive refractive power of the image capturing lens assembly can be balanced, and the photosensitivity and spherical aberration thereof can be reduced.

[0043] The sixth lens element can have an object-side surface being convex in a paraxial region thereof, and has an image-side surface being concave in a paraxial region thereof. Therefore, the principal point of the image capturing lens assembly can be shifted toward the object side, so that the back focal length and the total track length can be shortened effectively.

[0044] The seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, and the image-side surface of the seventh lens element includes at least one convex shape in an off-axis region thereof. Therefore, the principal point of the image capturing lens assembly can be shifted toward the object side, so that the back focal length and the total track length can be shortened effectively. Moreover, aberrations of the off-axis field can be corrected.

[0045] When an axial distance between an object-side surface of the first lens element and an image surface is TL, and an entrance pupil diameter of the image capturing lens assembly is EPD, the following condition is satisfied:

$1.0 < TL/EPD \leq 2.40$. Therefore, the features of large field of view and short total track length can be enhanced, and the brightness in the off-axis region of the image can be increased. Preferably, the following condition can be satisfied: $1.0 < TL/EPD \leq 2.10$. More preferably, the following condition can be satisfied: $1.20 < TL/EPD \leq 2.10$. More preferably, the following condition can be satisfied: $1.30 < TL/EPD \leq 2.0$.

[0046] When the axial distance between the object-side surface of the first lens element and the image surface is TL, and a maximum image height of the image capturing lens assembly is ImgH, the following condition can be satisfied: $0.80 < TL/ImgH < 1.70$. Therefore, the feature of compactness of the image capturing lens assembly can be enhanced.

[0047] When an f-number of the image capturing lens assembly is Fno, the following condition can be satisfied: $1.0 < Fno \leq 1.70$. Therefore, the image capturing lens assembly can be featured with a large aperture, and the functionality in depth of field photos can be improved.

[0048] When a vertical distance between a maximum effective diameter position on the image-side surface of the seventh lens element and an optical axis is Y72, and the entrance pupil diameter of the image capturing lens assembly is EPD, the following condition can be satisfied: $1.30 < 2 \times Y72/EPD < 2.30$. Therefore, it is favorable for balancing the large field of view and the volume of the image capturing lens assembly, and is favorable for the compactness of the image capturing lens assembly.

[0049] When an incident angle of a chief ray at the maximum image height on the image surface of the image capturing lens assembly is CRA1.0Y, the following condition can be satisfied: $32.0 \text{ degrees} \leq C.RA1.0Y$. Therefore, it is favorable for the coordination between the image capturing lens assembly and the image sensor, so that the image quality can be enhanced.

[0050] When an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, and an axial distance between the sixth lens element and the seventh lens element is T67, the following conditions can be satisfied: $1.0 < T67/T12$; $1.0 < T67/T23$; $1.0 < T67/T34$; $1.0 < T67/T45$; and $1.0 < T67/T56$. Therefore, the axial distance between the sixth lens element and the seventh lens element can be enlarged, so that the surface shapes of the sixth lens element and the seventh lens element can be properly arranged, which is favorable for enhancing the image quality in an off-axis region. Preferably, the following conditions can be satisfied: $2.0 < T67/T12 < 5.0$; $1.25 < T67/T23 < 3.0$; $1.0 < T67/T34 < 1.0$; $1.5 < T67/T45 < 1.5$; and $3.0 < T67/T56$. More preferably, the following conditions can be satisfied: $3.0 < T67/T12 < 3.0$; $1.75 < T67/T12 < 2.0$; $1.25 < T67/T34 < 5.0$; $2.0 < T67/T45 < 7.5$; and $5.0 < T67/T56 < 100$.

[0051] When the axial distance between the second lens element and the third lens element is T23, and the axial distance between the third lens element and the fourth lens element is T34, the following condition can be satisfied: $0 < T23/T34 \leq 0.90$. Therefore, the poor molding results generated from the excessive curvature of the second lens element and the third lens element can be prevented. Alternatively,

the improper refraction of light generated from the excessive curvature in the off-axis region of lens elements can be prevented.

[0052] When a focal length of the image capturing lens assembly is f, a curvature radius of the object-side surface of the seventh lens element is R13, and a curvature radius of the image-side surface of the seventh lens element is R14, the following condition can be satisfied: $4.0 < (f/R13) + (f/R14) < 10.0$. Therefore, the principal point of the image capturing lens assembly can be shifted toward the object side by the arrangement of the seventh lens element, which is favorable for reducing the back focal length.

[0053] When the entrance pupil diameter of the image capturing lens assembly is EPD, and the curvature radius of the image-side surface of the seventh lens element is R14, the following condition can be satisfied: $1.50 < EPD/R14 < 4.0$. Therefore, the principal point of the image capturing lens assembly can be further shifted toward the object side while reducing the back focal length.

[0054] When the axial distance between the object-side surface of the first lens element and the image surface is TL, the entrance pupil diameter of the image capturing lens assembly is EPD, and the maximum image height of the image capturing lens assembly is ImgH, the following condition can be satisfied: $2.0 < (TL^2)/(EPD \times ImgH) < 3.20$. Therefore, the feature of compactness can be further satisfied. Preferably, the following condition can be satisfied: $2.0 < (TL^2)/(EPD \times ImgH) \leq 3.0$.

[0055] When a total number of lens elements having Abbe numbers less than 40 is V40, the following condition can be satisfied: $4 \leq V40$. Therefore, the ability of the image capturing lens assembly for eliminating chromatic aberration can be improved, and the image quality can be maintained with the large aperture configuration.

[0056] When a total number of lens elements having Abbe numbers less than 30 is V30, the following condition can be satisfied: $3 \leq V30$. Therefore, the ability of the image capturing lens assembly for eliminating chromatic aberration can be further improved.

[0057] When a total number of lens elements having Abbe numbers less than 20 is V20, the following condition can be satisfied: $2 \leq V20$. Therefore, the ability of the image capturing lens assembly for eliminating chromatic aberration can be further improved.

[0058] Each of the aforementioned features of the image capturing lens assembly can be utilized in numerous combinations, so as to achieve the corresponding functionality.

[0059] According to the image capturing lens assembly of the present disclosure, the lens elements thereof can be made of a plastic material or a glass material. When the lens elements are made of a plastic material, the manufacturing cost can be effectively reduced. When the lens elements are made of a glass material, the arrangement of the refractive power of the image capturing lens assembly may be more flexible to design. Furthermore, surfaces of each lens element can be arranged to be aspheric (ASP), since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating aberrations thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the image capturing lens assembly can also be reduced.

[0060] According to the image capturing lens assembly of the present disclosure, each of an object-side surface and an

image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface away from the paraxial region. Particularly unless otherwise specified, when the lens element has a convex surface, it indicates that the surface can be convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface can be concave in the paraxial region thereof. According to the image capturing lens assembly of the present disclosure, the refractive power of a lens element being positive or negative or the focal length of the lens element may refer to the refractive power or the focal length in the paraxial region of the lens element.

[0061] According to the image capturing lens assembly of the present disclosure, the image capturing lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. The glare stop or the field stop is for eliminating the stray light and thereby improving the image resolution thereof.

[0062] According to the image capturing lens assembly of the present disclosure, the image surface, depending on the corresponding image sensor, can be a planar surface or a curved surface with any curvature, particularly a curved surface being concave toward the object side. According to the image capturing lens assembly of the present disclosure, at least one image correcting element (such as a field flattener) can be selectively disposed between a lens element closest to the image surface and the image surface so as to correct image aberrations (such as the field curvature). Properties of the image correcting element, such as curvature, thickness, refractive index, position, surface shape (convex/concave, spherical/aspheric/diffractive/Fresnel etc.) can be adjusted according to the requirements of the imaging apparatus. In general, the image correcting element is preferably a thin plano-concave element having a concave surface toward the object side and is disposed close to the image surface.

[0063] According to the image capturing lens assembly of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the image capturing lens assembly and the image surface to enable a telecentric effect, and thereby can improve the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image surface is favorable for enlarging the field of view of the image capturing lens assembly and thereby provides a wider field of view for the same.

[0064] According to the image capturing lens assembly of the present disclosure, the image capturing lens assembly can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, smart TVs, network monitoring devices, motion sensing input devices, driving recorders, rear view camera systems, wearable devices, unmanned aerial vehicles, and other electronic imaging products.

[0065] According to the present disclosure, an imaging apparatus is provided. The imaging apparatus includes the aforementioned image capturing lens assembly according to the present disclosure and an image sensor, wherein the image sensor is disposed on or near the image surface of the aforementioned image capturing lens assembly. With the image capturing lens assembly having seven lens elements,

the imaging functionality of in an off-axis region can be enhanced, which is more apparent under the configuration of large aperture. With the sixth lens element having an image-side surface being concave in a paraxial region thereof and the seventh lens element having an image-side surface being concave in a paraxial region thereof, the principal point of the image capturing lens assembly can be shifted toward the object side, so that the back focal length and the total track length can be shortened effectively. Accordingly, the image capturing lens assembly and the imaging apparatus according to the present disclosure become applicable to portable electronic devices. Preferably, the imaging apparatus can further include a barrel member, a holder member or a combination thereof.

[0066] According to the present disclosure, an electronic device is provided, wherein the electronic device includes the aforementioned imaging apparatus. Therefore, it is favorable for enhancing the image quality. Preferably, the electronic device can further include, but not limited to, a control unit, a display, a storage unit, a random access memory unit (RAM) or a combination thereof.

[0067] According to the above description of the present disclosure, the following 1st-14th specific embodiments are provided for further explanation.

1st Embodiment

[0068] FIG. 1 is a schematic view of an imaging apparatus according to the 1st embodiment of the present disclosure. FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 1st embodiment. In FIG. 1, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 195. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 110, an aperture stop 100, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, a seventh lens element 170, an IR-cut filter 180 and an image surface 190. The image sensor 195 is disposed on the image surface 190 of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (110, 120, 130, 140, 150, 160 and 170) without additional one or more lens elements inserted between the first lens element 110 and the seventh lens element 170.

[0069] The first lens element 110 with negative refractive power has an object-side surface 111 being concave in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof. The first lens element 110 is made of a plastic material, and has the object-side surface 111 and the image-side surface 112 being both aspheric.

[0070] The second lens element 120 with positive refractive power has an object-side surface 121 being convex in a paraxial region thereof and an image-side surface 122 being convex in a paraxial region thereof. The second lens element 120 is made of a plastic material, and has the object-side surface 121 and the image-side surface 122 being both aspheric.

[0071] The third lens element 130 with negative refractive power has an object-side surface 131 being convex in a paraxial region thereof and an image-side surface 132 being concave in a paraxial region thereof. The third lens element

130 is made of a plastic material, and has the object-side surface **131** and the image-side surface **132** being both aspheric.

[0072] The fourth lens element **140** with negative refractive power has an object-side surface **141** being convex in a paraxial region thereof and an image-side surface **142** being concave in a paraxial region thereof. The fourth lens element **140** is made of a plastic material, and has the object-side surface **141** and the image-side surface **142** being both aspheric.

[0073] The fifth lens element **150** with positive refractive power has an object-side surface **151** being concave in a paraxial region thereof and an image-side surface **152** being convex in a paraxial region thereof. The fifth lens element **150** is made of a plastic material, and has the object-side surface **151** and the image-side surface **152** being both aspheric.

[0074] The sixth lens element **160** with negative refractive power has an object-side surface **161** being convex in a paraxial region thereof and an image-side surface **162** being concave in a paraxial region thereof. The sixth lens element **160** is made of a plastic material, and has the object-side surface **161** and the image-side surface **162** being both aspheric.

[0075] The seventh lens element **170** with negative refractive power has an object-side surface **171** being convex in a paraxial region thereof and an image-side surface **172** being concave in a paraxial region thereof. The seventh lens element **170** is made of a plastic material, and has the object-side surface **171** and the image-side surface **172** being both aspheric. Moreover, the image-side surface **172** of the seventh lens element **170** includes at least one convex shape in an off-axis region thereof.

[0076] The IR-cut filter **180** is made of a glass material and located between the seventh lens element **170** and the image surface **190**, and will not affect the focal length of the image capturing lens assembly.

[0077] The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1 + k) \times (Y/R)^2}) + \sum_i (A_i) \times (Y^i),$$

[0078] where,

[0079] X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

[0080] Y is the vertical distance from the point on the aspheric surface to the optical axis;

[0081] R is the curvature radius;

[0082] k is the conic coefficient; and

[0083] A_i is the i -th aspheric coefficient.

[0084] In the image capturing lens assembly according to the 1st embodiment, when a focal length of the image capturing lens assembly is f , an f -number of the image capturing lens assembly is Fno , and half of a maximum field of view of the image capturing lens assembly is $HFOV$, these parameters have the following values: $f=4.36$ mm; $Fno=1.75$; and $HFOV=38.6$ degrees.

[0085] FIG. 21 is a schematic view showing a parameter of $CRA1.0Y$ of the imaging apparatus according to the 1st

embodiment in FIG. 1. As shown in FIG. 21, when an incident angle of a chief ray at a maximum image height on the image surface **190** of the image capturing lens assembly is $CRA1.0Y$, the following condition is satisfied: $CRA1.0Y=34.9$ degrees.

[0086] In the image capturing lens assembly according to the 1st embodiment, when a total number of lens elements having Abbe numbers less than 40 is $V40$, the following condition is satisfied: $V40=4$. Specifically, in the image capturing lens assembly according to the 1st embodiment, the lens elements having Abbe numbers less than 40 are the first lens element **110** (Abbe number=26.0), the third lens element **130** (Abbe number=26.0), the fourth lens element **140** (Abbe number=21.5) and the sixth lens element **160** (Abbe number=23.5).

[0087] In the image capturing lens assembly according to the 1st embodiment, when a total number of lens elements having Abbe numbers less than 30 is $V30$, the following condition is satisfied: $V30=4$. Specifically, in the image capturing lens assembly according to the 1st embodiment, the lens elements having Abbe numbers less than 30 are the first lens element **110** (Abbe number=26.0), the third lens element **130** (Abbe number=26.0), the fourth lens element **140** (Abbe number=21.5) and the sixth lens element **160** (Abbe number=23.5).

[0088] In the image capturing lens assembly according to the 1st embodiment, when a total number of lens elements having Abbe numbers less than 20 is $V20$, the following condition is satisfied: $V20=0$. Specifically, in the image capturing lens assembly according to the 1st embodiment, each of the first lens element **110** (Abbe number=26.0), the second lens element **120** (Abbe number=56.0), the third lens element **130** (Abbe number=26.0), the fourth lens element **140** (Abbe number=21.5), the fifth lens element **150** (Abbe number=56.0), the sixth lens element **160** (Abbe number=23.5) and the seventh lens element **170** (Abbe number=56.0) has an Abbe number greater than 20.

[0089] In the image capturing lens assembly according to the 1st embodiment, when an axial distance between the first lens element **110** and the second lens element **120** is $T12$, an axial distance between the second lens element **120** and the third lens element **130** is $T23$, an axial distance between the third lens element **130** and the fourth lens element **140** is $T34$, an axial distance between the fourth lens element **140** and the fifth lens element **150** is $T45$, an axial distance between the fifth lens element **150** and the sixth lens element **160** is $T56$, and an axial distance between the sixth lens element **160** and the seventh lens element **170** is $T67$, the following conditions are satisfied: $T23/T34=0.12$; $T67/T12=13.83$; $T67/T23=14.66$; $T67/T34=1.75$; $T67/T45=2.77$; and $T67/T56=14.66$.

[0090] In the image capturing lens assembly according to the 1st embodiment, when an axial distance between the object-side surface **111** of the first lens element **110** and the image surface **190** is TL , an entrance pupil diameter of the image capturing lens assembly is EPD , and the maximum image height of the image capturing lens assembly is $ImgH$ (half of a diagonal length of an effective photosensitive area of the image sensor **195**), the following conditions are satisfied: $TL/EPD=2.35$; $TL/ImgH=1.66$; and $(TL^2)/(EPD \times ImgH)=3.90$.

[0091] In the image capturing lens assembly according to the 1st embodiment, when the entrance pupil diameter of the image capturing lens assembly is EPD , and a curvature

radius of the image-side surface **172** of the seventh lens element **170** is R14, the following condition is satisfied: $EPD/R14=1.36$.

[0092] FIG. 22 is a schematic view showing a parameter of Y72 of the imaging apparatus according to the 1st embodiment in FIG. 1. As shown in FIG. 22, when a vertical distance between a maximum effective diameter position on the image-side surface **172** of the seventh lens element **170** and the optical axis is Y72, and the entrance pupil diameter of the image capturing lens assembly is EPD, the following condition is satisfied: $2 \times Y72/EPD=2.20$.

[0093] In the image capturing lens assembly according to the 1st embodiment, when the focal length of the image capturing lens assembly is f, a curvature radius of the object-side surface **171** of the seventh lens element **170** is R13, and the curvature radius of the image-side surface **172** of the seventh lens element **170** is R14, the following condition is satisfied: $(f/R13)+(f/R14)=3.07$.

[0094] The detailed optical data of the 1st embodiment are shown in TABLE 1 and the aspheric surface data are shown in TABLE 2 below.

TABLE 1

1st Embodiment								
f = 4.36 mm, Fno = 1.75, HFOV = 38.6 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	-33.630 ASP	0.240	Plastic	1.614	26.0	-37.61	
2		73.790 ASP	0.454					
3	Ape. Stop	Plano	-0.401					
4	Lens 2	1.987 ASP	0.812	Plastic	1.544	56.0	3.39	
5		-22.270 ASP	0.050					
6	Lens 3	146.754 ASP	0.286	Plastic	1.614	26.0	-10.17	
7		5.983 ASP	0.420					
8	Lens 4	12.570 ASP	0.431	Plastic	1.650	21.5	-34.39	
9		7.939 ASP	0.265					
10	Lens 5	-3.576 ASP	0.757	Plastic	1.544	56.0	3.99	
11		-1.452 ASP	0.050					
12	Lens 6	3.349 ASP	0.350	Plastic	1.639	23.5	-12.11	
13		2.242 ASP	0.733					
14	Lens 7	6.228 ASP	0.370	Plastic	1.544	56.0	-4.94	
15		1.838 ASP	0.500					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.328					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 7 is 1.150 mm.

TABLE 2

Aspheric Coefficients								
	Surface #							
	1	2	4	5	6	7	8	
k =	1.3483E+01	5.6949E+01	-2.9566E-01	-1.0000E+00	-9.0000E+01	-7.5095E+01	-8.0848E+01	
A4 =	-3.7828E-03	-3.7066E-02	-4.1674E-02	-1.0459E-01	-1.1790E-01	-1.9239E-02	-1.5640E-01	
A6 =	-1.5998E-02	2.4347E-02	3.6732E-02	8.1805E-02	1.7205E-01	8.3617E-02	1.6286E-01	
A8 =	2.6880E-02	8.7162E-03	-2.3112E-02	1.0052E-02	-4.9012E-02	-8.3411E-02	-4.5355E-01	
A10 =	-1.4272E-02	-9.9692E-03	9.1558E-03	-6.2670E-02	-5.6396E-02	6.0572E-02	7.1451E-01	
A12 =	3.5159E-03	3.8310E-03	-1.6636E-03	4.0853E-02	4.9735E-02	-4.3838E-02	-6.4976E-01	
A14 =	-3.7281E-04	-6.7041E-04		-8.6984E-03	-1.1230E-02	1.4752E-02	3.0088E-01	
A16 =							-5.2769E-02	
	Surface #							
	9	10	11	12	13	14	15	
k =	-1.7616E+01	-4.3378E+00	-1.0924E+00	1.0170E+00	-1.1621E+00	-2.2775E+00	-9.8730E-01	
A4 =	-1.1679E-01	-4.0101E-02	4.4007E-02	-5.2206E-02	-9.5410E-02	-1.4295E-01	-1.7644E-01	
A6 =	1.1778E-01	8.8954E-02	-5.2955E-02	-2.2722E-02	1.9234E-02	2.4666E-03	5.9200E-02	
A8 =	-2.1121E-01	-1.1389E-01	5.6194E-02	2.1008E-02	-4.9241E-04	1.8582E-02	-1.6343E-02	
A10 =	2.3045E-01	6.2652E-02	-4.4095E-02	-1.1658E-02	-2.6120E-03	-6.3756E-03	3.2704E-03	
A12 =	-1.4895E-01	-1.1347E-02	1.9102E-02	2.8249E-03	1.0030E-03	9.9766E-04	-4.0929E-04	
A14 =	5.3730E-02		-3.3225E-03	-2.3423E-04	-1.4591E-04	-7.8320E-05	2.7669E-05	
A16 =	-7.8451E-03		1.2685E-04		7.6330E-06	2.5034E-06	-7.6699E-07	

[0095] In TABLE 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-18 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In TABLE 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as TABLE 1 and TABLE 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

[0096] FIG. 3 is a schematic view of an imaging apparatus according to the 2nd embodiment of the present disclosure. FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 2nd embodiment. In FIG. 3, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 295. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 210, an aperture stop 200, a second lens element 220, a third lens element 230, a fourth lens element 240, a fifth lens element 250, a sixth lens element 260, a seventh lens element 270, an IR-cut filter 280 and an image surface 290. The image sensor 295 is disposed on the image surface 290 of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (210, 220, 230, 240, 250, 260 and 270) without additional one or more lens elements inserted between the first lens element 210 and the seventh lens element 270.

[0097] The first lens element 210 with negative refractive power has an object-side surface 211 being convex in a paraxial region thereof and an image-side surface 212 being concave in a paraxial region thereof. The first lens element 210 is made of a plastic material, and has the object-side surface 211 and the image-side surface 212 being both aspheric.

[0098] The second lens element 220 with positive refractive power has an object-side surface 221 being convex in a paraxial region thereof and an image-side surface 222 being convex in a paraxial region thereof. The second lens element

220 is made of a plastic material, and has the object-side surface 221 and the image-side surface 222 being both aspheric.

[0099] The third lens element 230 with negative refractive power has an object-side surface 231 being concave in a paraxial region thereof and an image-side surface 232 being concave in a paraxial region thereof. The third lens element 230 is made of a plastic material, and has the object-side surface 231 and the image-side surface 232 being both aspheric.

[0100] The fourth lens element 240 with negative refractive power has an object-side surface 241 being convex in a paraxial region thereof and an image-side surface 242 being concave in a paraxial region thereof. The fourth lens element 240 is made of a plastic material, and has the object-side surface 241 and the image-side surface 242 being both aspheric.

[0101] The fifth lens element 250 with positive refractive power has an object-side surface 251 being concave in a paraxial region thereof and an image-side surface 252 being convex in a paraxial region thereof. The fifth lens element 250 is made of a plastic material, and has the object-side surface 251 and the image-side surface 252 being both aspheric.

[0102] The sixth lens element 260 with negative refractive power has an object-side surface 261 being convex in a paraxial region thereof and an image-side surface 262 being concave in a paraxial region thereof. The sixth lens element 260 is made of a plastic material, and has the object-side surface 261 and the image-side surface 262 being both aspheric.

[0103] The seventh lens element 270 with negative refractive power has an object-side surface 271 being convex in a paraxial region thereof and an image-side surface 272 being concave in a paraxial region thereof. The seventh lens element 270 is made of a plastic material, and has the object-side surface 271 and the image-side surface 272 being both aspheric. Moreover, the image-side surface 272 of the seventh lens element 270 includes at least one convex shape in an off-axis region thereof.

[0104] The IR-cut filter 280 is made of a glass material and located between the seventh lens element 270 and the image surface 290, and will not affect the focal length of the image capturing lens assembly.

[0105] The detailed optical data of the 2nd embodiment are shown in TABLE 3 and the aspheric surface data are shown in TABLE 4 below.

TABLE 3

2nd Embodiment								
f = 3.85 mm, Fno = 1.68, HFOV = 43.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	3.779 ASP	0.350	Plastic	1.603	28.5	-11.44	
2		2.356 ASP	0.379					
3	Ape. Stop	Plano	-0.300					
4	Lens 2	1.751 ASP	0.660	Plastic	1.544	55.9	2.72	
5		-8.227 ASP	0.050					
6	Lens 3	-95.285 ASP	0.236		1.584	28.2	-10.29	
7		6.421 ASP	0.392					
8	Lens 4	6.655 ASP	0.268	Plastic	1.669	19.5	-31.38	
9		4.972 ASP	0.262					
10	Lens 5	-4.111 ASP	0.714		1.544	55.9	4.05	
11		-1.522 ASP	0.050					
12	Lens 6	3.687 ASP	0.350	Plastic	1.669	19.5	-24.03	

TABLE 3-continued

2nd Embodiment f = 3.85 mm, Fno = 1.68, HFOV = 43.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
13		2.885	ASP	0.741				
14	Lens 7	3.321	ASP	0.370	Plastic	1.544	55.9	-4.59
15		1.370	ASP	0.300				
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.416					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 7 is 1.150 mm.

TABLE 4

Aspheric Coefficients							
	Surface #						
	1	2	4	5	6	7	8
k =	-1.1832E-02	-2.8460E+00	-1.2663E+00	-8.5581E+01	9.0000E+01	-6.7906E+01	-8.5892E+01
A4 =	-4.1736E-02	-1.2985E-01	-1.1033E-01	-7.7399E-02	-5.2046E-02	-2.0490E-02	-1.8744E-01
A6 =	1.8169E-02	1.2839E-01	1.0797E-01	9.3741E-02	1.7952E-01	1.1868E-01	2.3088E-01
A8 =	-1.1329E-02	-6.5451E-02	-8.0246E-02	-8.4426E-02	-1.5553E-01	-1.8256E-01	-7.5285E-01
A10 =	7.2727E-03	1.8842E-02	4.6488E-02	3.0745E-02	5.4456E-02	1.6948E-01	1.2464E+00
A12 =	-2.9688E-03	6.2100E-03	-1.0580E-02	9.5701E-03	1.0034E-02	-1.0509E-01	-1.1716E+00
A14 =	3.8863E-04	-4.5619E-03		-6.6896E-03	-9.5752E-03	2.7778E-02	5.7318E-01
A16 =							-1.0891E-01

	Surface #						
	9	10	11	12	13	14	15
k =	-5.6703E+00	-4.1083E+00	-9.0339E-01	-7.6196E+01	-6.3738E-02	-1.4319E+00	-2.6409E+00
A4 =	-1.8476E-01	-9.5215E-02	-8.4679E-02	8.3328E-02	-3.4205E-02	-2.0385E-01	-1.6552E-01
A6 =	2.2204E-01	1.9771E-01	2.5379E-01	-8.4535E-02	-1.1016E-02	5.4134E-02	7.9844E-02
A8 =	-4.3750E-01	-2.3022E-01	-3.7070E-01	3.4577E-02	4.5891E-03	-1.1112E-02	-2.7876E-02
A10 =	5.0265E-01	1.1735E-01	3.1428E-01	-9.1535E-03	-1.4026E-03	2.5722E-03	5.9289E-03
A12 =	-3.4037E-01	-2.0755E-02	-1.5951E-01	1.2269E-03	2.9046E-04	-4.2594E-04	-7.2720E-04
A14 =	1.2956E-01		4.4924E-02	-5.5112E-05	-3.0929E-05	3.6843E-05	4.6554E-05
A16 =	-2.0234E-02		-5.2191E-03		1.2453E-06	-1.2511E-06	-1.1769E-06

[0106] In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

[0107] Moreover, these parameters can be calculated from TABLE 3 and TABLE 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f (mm)	3.85	T67/T34	1.89
Fno	1.68	T67/T45	2.83
HFOV (deg.)	43.0	T67/T56	14.82
CRA1.0Y (deg.)	31.9	TL/EPD	2.38
V40	4	TL/ImgH	1.54
V30	4	(TL ²)/(EPD × ImgH)	3.67
V20	2	EPD/R14	1.67
T23/T34	0.13	2 × Y72/EPD	2.41

-continued

2nd Embodiment			
T67/T12	9.38	f/R13 + f/R14	3.97
T67/T23	14.82		

3rd Embodiment

[0108] FIG. 5 is a schematic view of an imaging apparatus according to the 3rd embodiment of the present disclosure. FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 3rd embodiment. In FIG. 5, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 395. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 300, a first lens element 310, a second lens element 320, a third lens element 330, a stop 301, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, a seventh lens element 370, an IR-cut filter 380 and an image surface 390. The image sensor 395 is disposed on the image surface 390.

of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (310, 320, 330, 340, 350, 360 and 370) without additional one or more lens elements inserted between the first lens element 310 and the seventh lens element 370.

[0109] The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being concave in a paraxial region thereof. The first lens element 310 is made of a plastic material, and has the object-side surface 311 and the image-side surface 312 being both aspheric.

[0110] The second lens element 320 with negative refractive power has an object-side surface 321 being convex in a paraxial region thereof and an image-side surface 322 being concave in a paraxial region thereof. The second lens element 320 is made of a plastic material, and has the object-side surface 321 and the image-side surface 322 being both aspheric.

[0111] The third lens element 330 with positive refractive power has an object-side surface 331 being convex in a paraxial region thereof and an image-side surface 332 being concave in a paraxial region thereof. The third lens element 330 is made of a plastic material, and has the object-side surface 331 and the image-side surface 332 being both aspheric.

[0112] The fourth lens element 340 with negative refractive power has an object-side surface 341 being concave in a paraxial region thereof and an image-side surface 342 being concave in a paraxial region thereof. The fourth lens

element 340 is made of a plastic material, and has the object-side surface 341 and the image-side surface 342 being both aspheric.

[0113] The fifth lens element 350 with positive refractive power has an object-side surface 351 being concave in a paraxial region thereof and an image-side surface 352 being convex in a paraxial region thereof. The fifth lens element 350 is made of a plastic material, and has the object-side surface 351 and the image-side surface 352 being both aspheric.

[0114] The sixth lens element 360 with negative refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a paraxial region thereof. The sixth lens element 360 is made of a plastic material, and has the object-side surface 361 and the image-side surface 362 being both aspheric.

[0115] The seventh lens element 370 with negative refractive power has an object-side surface 371 being convex in a paraxial region thereof and an image-side surface 372 being concave in a paraxial region thereof. The seventh lens element 370 is made of a plastic material, and has the object-side surface 371 and the image-side surface 372 being both aspheric. Moreover, the image-side surface 372 of the seventh lens element 370 includes at least one convex shape in an off-axis region thereof.

[0116] The IR-cut filter 380 is made of a glass material and located between the seventh lens element 370 and the image surface 390, and will not affect the focal length of the image capturing lens assembly.

[0117] The detailed optical data of the 3rd embodiment are shown in TABLE 5 and the aspheric surface data are shown in TABLE 6 below.

TABLE 5

3rd Embodiment									
f = 3.93 mm, Fno = 1.45, HFOV = 39.0 deg.									
Surface #		Curvature Radius		Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano		Infinity					
1	Ape. Stop	Plano		-0.385					
2	Lens 1	1.831	ASP	0.656	Plastic	1.545	56.0	6.49	
3		3.321	ASP	0.069					
4	Lens 2	2.725	ASP	0.230	Plastic	1.671	19.5	-11.54	
5		1.947	ASP	0.157					
6	Lens 3	2.066	ASP	0.464	Plastic	1.544	56.0	5.63	
7		5.839	ASP	0.022					
8	Stop	Plano		0.308					
9	Lens 4	-200.000	ASP	0.331	Plastic	1.671	19.5	-14.11	
10		9.942	ASP	0.103					
11	Lens 5	-4.251	ASP	0.439	Plastic	1.544	56.0	4.59	
12		-1.631	ASP	0.050					
13	Lens 6	4.561	ASP	0.342	Plastic	1.671	19.5	-26.37	
14		3.517	ASP	0.385					
15	Lens 7	2.220	ASP	0.360	Plastic	1.544	56.0	-5.32	
16		1.184	ASP	0.500					
17	IR-cut filter	Plano		0.210	Glass	1.517	64.2	—	
18		Plano		0.357					
19	Image	Plano		—					

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 8 is 1.120 mm.

TABLE 6

Aspheric Coefficients							
Surface #	2	3	4	5	6	7	9
k =	-1.6163E-01	-2.0738E+00	-2.8201E+01	-1.1493E+01	-9.3281E+00	9.7527E+00	7.9638E+01
A4 =	-4.0064E-03	-5.1562E-02	1.5588E-02	7.7222E-03	4.9275E-02	-3.1363E-02	-1.7890E-01
A6 =	4.9611E-03	-7.4285E-02	-1.8317E-01	-7.3507E-02	-1.0971E-01	-2.5699E-02	8.5434E-02
A8 =	-2.8885E-02	7.7688E-02	3.0353E-01	1.8107E-01	1.1549E-01	-1.0024E-02	-2.3749E-01
A10 =	3.0769E-02	-3.4539E-02	-2.1953E-01	-1.2866E-01	-8.9566E-02	-1.4398E-02	3.0298E-01
A12 =	-2.1166E-02	5.3315E-03	8.8189E-02	3.6529E-02	2.0915E-02	1.0149E-02	-1.3981E-01
A14 =	4.5109E-03		-1.5311E-02	1.3780E-03	7.3782E-03		-7.1273E-03
A16 =							1.5042E-02
Surface #	10	11	12	13	14	15	16
k =	5.3352E+01	9.0120E+00	-1.5865E+00	1.2182E+00	-6.2708E+00	-1.1859E+01	-5.2863E+00
A4 =	-3.4223E-01	-2.6215E-01	1.6853E-01	2.3436E-01	7.5220E-02	-3.7524E-01	-2.4278E-01
A6 =	9.4020E-01	1.4875E+00	-2.9006E-01	-6.2355E-01	-1.9860E-01	2.6902E-01	1.8159E-01
A8 =	-2.0272E+00	-3.0293E+00	3.9149E-01	7.3043E-01	1.4789E-01	-2.0915E-01	-1.1540E-01
A10 =	2.2808E+00	3.2050E+00	-3.5355E-01	-5.9388E-01	-7.0411E-02	1.3967E-01	5.4021E-02
A12 =	-1.3712E+00	-1.8613E+00	1.8770E-01	3.2195E-01	2.0537E-02	-5.9221E-02	-1.6841E-02
A14 =	4.1306E-01	5.5797E-01	-5.1124E-02	-1.1327E-01	-3.1935E-03	1.5325E-02	3.3678E-03
A16 =	-4.8394E-02	-6.7223E-02	5.4433E-03	2.3361E-02	2.0117E-04	-2.3800E-03	-4.1595E-04
A18 =				-2.0944E-03		2.0525E-04	2.9004E-05
A20 =						-7.5976E-06	-8.7710E-07

[0118] In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

[0119] Moreover, these parameters can be calculated from TABLE 5 and TABLE 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f (mm)	3.93	T67/T34	1.17
Fno	1.45	T67/T45	3.74
HFOV (deg.)	39.0	T67/T56	7.70
CRA1.0Y (deg.)	35.9	TL/EPD	1.84
V40	3	TL/ImgH	1.54
V30	3	(TL ²)/(EPD × ImgH)	2.83
V20	3	EPD/R14	2.29
T23/T34	0.48	2 × Y72/EPD	1.81
T67/T12	5.58	f/R13 + f/R14	5.09
T67/T23	2.45		

4th Embodiment

[0120] FIG. 7 is a schematic view of an imaging apparatus according to the 4th embodiment of the present disclosure. FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 4th embodiment. In FIG. 7, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 495. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a stop 401, a third lens element 430, a stop 402, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, a seventh lens element 470, an IR-cut filter 480 and an image surface

490. The image sensor 495 is disposed on the image surface 490 of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (410, 420, 430, 440, 450, 460 and 470) without additional one or more lens elements inserted between the first lens element 410 and the seventh lens element 470.

[0121] The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof. The first lens element 410 is made of a plastic material, and has the object-side surface 411 and the image-side surface 412 being both aspheric.

[0122] The second lens element 420 with negative refractive power has an object-side surface 421 being convex in a paraxial region thereof and an image-side surface 422 being concave in a paraxial region thereof. The second lens element 420 is made of a plastic material, and has the object-side surface 421 and the image-side surface 422 being both aspheric.

[0123] The third lens element 430 with positive refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof. The third lens element 430 is made of a plastic material, and has the object-side surface 431 and the image-side surface 432 being both aspheric.

[0124] The fourth lens element 440 with negative refractive power has an object-side surface 441 being concave in a paraxial region thereof and an image-side surface 442 being concave in a paraxial region thereof. The fourth lens element 440 is made of a plastic material, and has the object-side surface 441 and the image-side surface 442 being both aspheric.

[0125] The fifth lens element 450 with positive refractive power has an object-side surface 451 being concave in a paraxial region thereof and an image-side surface 452 being convex in a paraxial region thereof. The fifth lens element

450 is made of a plastic material, and has the object-side surface **451** and the image-side surface **452** being both aspheric.

[0126] The sixth lens element **460** with negative refractive power has an object-side surface **461** being convex in a paraxial region thereof and an image-side surface **462** being concave in a paraxial region thereof. The sixth lens element **460** is made of a plastic material, and has the object-side surface **461** and the image-side surface **462** being both aspheric.

[0127] The seventh lens element **470** with negative refractive power has an object-side surface **471** being convex in a paraxial region thereof and an image-side surface **472** being

concave in a paraxial region thereof. The seventh lens element **470** is made of a plastic material, and has the object-side surface **471** and the image-side surface **472** being both aspheric. Moreover, the image-side surface **472** of the seventh lens element **470** includes at least one convex shape in an off-axis region thereof.

[0128] The IR-cut filter **480** is made of a glass material and located between the seventh lens element **470** and the image surface **490**, and will not affect the focal length of the image capturing lens assembly.

[0129] The detailed optical data of the 4th embodiment are shown in TABLE 7 and the aspheric surface data are shown in TABLE 8 below.

TABLE 7

4th Embodiment							
f = 3.95 mm, Fno = 1.54, HFOV = 39.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe#	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.447				
2	Lens 1	1.804 ASP	0.682	Plastic	1.545	56.0	5.46
3		3.977 ASP	0.112				
4	Lens 2	2.783 ASP	0.230	Plastic	1.669	19.5	-14.96
5		2.105 ASP	0.209				
6	Stop	Plano	-0.009				
7	Lens 3	3.083 ASP	0.426	Plastic	1.544	56.0	7.83
8		10.621 ASP	-0.113				
9	Stop	Plano	0.359				
10	Lens 4	-64.351 ASP	0.299	Plastic	1.669	19.5	-12.69
11		9.797 ASP	0.112				
12	Lens 5	-4.045 ASP	0.444	Plastic	1.566	37.4	4.18
13		-1.551 ASP	0.050				
14	Lens 6	5.019 ASP	0.355	Plastic	1.669	19.5	-27.87
15		3.842 ASP	0.409				
16	Lens 7	2.962 ASP	0.360	Plastic	1.566	37.4	-4.39
17		1.292 ASP	0.500				
18	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
19		Plano	0.335				
20	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 6 is 1.055 mm.

The effective radius of the surface 9 is 1.120 mm.

TABLE 8

Aspheric Coefficients							
Surface #	2	3	4	5	7	8	10
k =	-3.6818E-01	-3.8570E+00	-3.7508E+01	-1.9429E+01	-2.9332E+01	-8.1557E+00	3.8601E+01
A4 =	-1.9662E-03	-1.3385E-01	-9.1973E-02	-3.9305E-03	3.5559E-02	-6.3602E-02	-2.2413E-01
A6 =	2.0708E-02	1.4436E-01	-2.7913E-02	-7.6238E-02	-1.0295E-01	-4.0777E-02	4.3960E-02
A8 =	-3.5051E-02	-1.2712E-01	2.6757E-01	1.6477E-01	-1.9121E-02	-2.4391E-02	-9.0078E-02
A10 =	2.8670E-02	5.7411E-02	-3.5815E-01	-6.7234E-02	1.3003E-01	9.1403E-03	-4.0909E-02
A12 =	-1.2548E-02	-1.1526E-02	2.2179E-01	-6.3460E-02	-1.8002E-01	3.1218E-03	4.1421E-01
A14 =	8.7461E-04	-3.8881E-05	-5.1797E-02	5.5806E-02	8.5098E-02		-4.3064E-01
A16 =							1.3217E-01
Surface #	11	12	13	14	15	16	17
k =	-1.7194E+01	8.3674E+00	-1.1430E+00	5.4988E+00	-1.6812E+00	-2.5544E+01	-6.4772E+00
A4 =	-3.6789E-01	-2.4503E-01	2.1539E-01	3.4843E-01	1.6540E-01	-3.3492E-01	-2.3152E-01
A6 =	9.8444E-01	1.4528E+00	-5.5150E-01	-1.1582E+00	-4.8687E-01	2.3132E-01	1.6817E-01
A8 =	-2.3448E+00	-3.2608E+00	9.6831E-01	1.8174E+00	5.5046E-01	-2.1536E-01	-1.1175E-01
A10 =	2.8866E+00	3.7590E+00	-1.0182E+00	-1.9256E+00	-4.1065E-01	1.7357E-01	5.7153E-02
A12 =	-1.8532E+00	-2.3786E+00	6.0104E-01	1.3478E+00	2.0844E-01	-8.3338E-02	-1.9928E-02
A14 =	5.8466E-01	7.8649E-01	-1.8120E-01	-6.0271E-01	-7.0879E-02	2.3544E-02	4.5208E-03
A16 =	-7.0402E-02	-1.0610E-01	2.1713E-02	1.5910E-01	1.5378E-02	-3.8972E-03	-6.3681E-04
A18 =				-2.1217E-02	-1.9087E-03	3.5143E-04	5.0388E-05
A20 =				9.6741E-04	1.0237E-04	-1.3363E-05	-1.7038E-06

[0130] In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

[0131] Moreover, these parameters can be calculated from TABLE 7 and TABLE 8 as the following values and satisfy the following conditions:

4th Embodiment			
f (mm)	3.95	T67/T34	1.66
Fno	1.54	T67/T45	3.65
HFOV (deg.)	39.0	T67/T56	8.18
CRA1.0Y (deg.)	35.5	TL/EPD	1.94
V40	5	TL/ImgH	1.53
V30	3	(TL ²)/(EPD × ImgH)	2.97
V20	3	EPD/R14	1.99
T23/T34	0.81	2 × Y72/EPD	1.87
T67/T12	3.65	ƒR13 + ƒR14	4.39
T67/T23	2.05		

5th Embodiment

[0132] FIG. 9 is a schematic view of an imaging apparatus according to the 5th embodiment of the present disclosure. FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 5th embodiment. In FIG. 9, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 595. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, a seventh lens element 570, an IR-cut filter 580 and an image surface 590. The image sensor 595 is disposed on the image surface 590 of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (510, 520, 530, 540, 550, 560 and 570) without additional one or more lens elements inserted between the first lens element 510 and the seventh lens element 570.

[0133] The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being concave in a paraxial region thereof. The first lens element 510 is made of a plastic material, and has the object-side surface 511 and the image-side surface 512 being both aspheric.

[0134] The second lens element 520 with negative refractive power has an object-side surface 521 being convex in a paraxial region thereof and an image-side surface 522 being concave in a paraxial region thereof. The second lens element 520 is made of a plastic material, and has the object-side surface 521 and the image-side surface 522 being both aspheric.

[0135] The third lens element 530 with positive refractive power has an object-side surface 531 being convex in a paraxial region thereof and an image-side surface 532 being concave in a paraxial region thereof. The third lens element 530 is made of a plastic material, and has the object-side surface 531 and the image-side surface 532 being both aspheric.

[0136] The fourth lens element 540 with negative refractive power has an object-side surface 541 being convex in a paraxial region thereof and an image-side surface 542 being concave in a paraxial region thereof. The fourth lens element 540 is made of a plastic material, and has the object-side surface 541 and the image-side surface 542 being both aspheric.

[0137] The fifth lens element 550 with positive refractive power has an object-side surface 551 being concave in a paraxial region thereof and an image-side surface 552 being convex in a paraxial region thereof. The fifth lens element 550 is made of a plastic material, and has the object-side surface 551 and the image-side surface 552 being both aspheric.

[0138] The sixth lens element 560 with negative refractive power has an object-side surface 561 being convex in a paraxial region thereof and an image-side surface 562 being concave in a paraxial region thereof. The sixth lens element 560 is made of a plastic material, and has the object-side surface 561 and the image-side surface 562 being both aspheric.

[0139] The seventh lens element 570 with negative refractive power has an object-side surface 571 being convex in a paraxial region thereof and an image-side surface 572 being concave in a paraxial region thereof. The seventh lens element 570 is made of a plastic material, and has the object-side surface 571 and the image-side surface 572 being both aspheric. Moreover, the image-side surface 572 of the seventh lens element 570 includes at least one convex shape in an off-axis region thereof.

[0140] The IR-cut filter 580 is made of a glass material and located between the seventh lens element 570 and the image surface 590, and will not affect the focal length of the image capturing lens assembly.

[0141] The detailed optical data of the 5th embodiment are shown in TABLE 9 and the aspheric surface data are shown in TABLE 10 below.

TABLE 9

5th Embodiment								
f = 3.94 mm, Fno = 1.70, HFOV = 38.6 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe#	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.411					
2	Lens 1	1.737 ASP	0.598	Plastic	1.545	56.0	4.05	

TABLE 9-continued

5th Embodiment								
f = 3.94 mm, Fno = 1.70, HFOV = 38.6 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe#	Focal Length	
3		7.178 ASP	0.050					
4	Lens 2	4.907 ASP	0.230	Plastic	1.660	20.4	-9.46	
5		2.697 ASP	0.242					
6	Lens 3	3.474 ASP	0.293	Plastic	1.544	56.0	20.40	
7		4.906 ASP	0.246					
8	Lens 4	9.195 ASP	0.448	Plastic	1.660	20.4	-73.62	
9		7.582 ASP	0.159					
10	Lens 5	-2.959 ASP	0.546	Plastic	1.547	52.9	4.70	
11		-1.466 ASP	0.050					
12	Lens 6	3.629 ASP	0.350	Plastic	1.642	21.3	-13.75	
13		2.475 ASP	0.350					
14	Lens 7	1.810 ASP	0.360	Plastic	1.544	56.0	-6.67	
15		1.123 ASP	0.500					
16	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
17		Plano	0.388					
18	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 8 is 1.150 mm.

TABLE 10

Aspheric Coefficients							
Surface #	2	3	4	5	6	7	8
k =	1.0042E-01	-2.6892E+01	-9.0000E+01	-1.6466E+01	-2.3238E+01	-9.0000E+01	3.4909E+01
A4 =	-5.4966E-03	-2.7792E-02	1.0060E-02	4.4515E-02	2.1767E-03	4.5801E-03	-1.5161E-01
A6 =	2.2588E-02	5.8394E-02	-3.0093E-02	-6.9434E-02	9.2372E-03	-1.3826E-02	3.3927E-02
A8 =	-4.8802E-02	-1.1380E-01	-1.1833E-03	5.5014E-02	-1.6688E-01	-9.5541E-02	-4.3321E-02
A10 =	5.0190E-02	7.7993E-02	6.8447E-03	-2.7778E-02	1.0660E-01	5.9207E-02	4.6448E-03
A12 =	-2.3903E-02	-2.1053E-02	2.4728E-02	3.1595E-02			4.2287E-03
A14 =			-9.9454E-03				3.9429E-03
Surface #	9	10	11	12	13	14	15
k =	-7.5249E+01	-3.7160E-01	-1.1327E+00	1.7314E+00	-4.2107E+00	-1.4495E+01	-5.5865E+00
A4 =	-1.0668E-01	-3.9500E-02	-4.6900E-02	-1.0069E-01	-8.9882E-02	-2.2346E-01	-1.6340E-01
A6 =	5.3195E-02	1.1021E-01	1.3889E-01	6.4311E-02	4.9731E-02	7.8700E-02	8.4378E-02
A8 =	-5.9052E-02	-7.6836E-02	-1.4182E-01	-7.5050E-02	-3.1373E-02	-1.2597E-02	-3.5495E-02
A10 =	2.9796E-02	2.3148E-02	9.2356E-02	4.2233E-02	1.0617E-02	1.4587E-03	1.0245E-02
A12 =	-4.6891E-03	9.9355E-04	-2.7016E-02	-1.5854E-02	-1.8922E-03	-2.2403E-04	-1.8070E-03
A14 =	-3.0021E-03	-1.2062E-03	2.1408E-03	3.4273E-03	1.8001E-04	2.8918E-05	1.7223E-04
A16 =	1.9547E-03		1.9840E-04	-2.8306E-04	-8.0060E-06	-1.5088E-06	-6.6932E-06

[0142] In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

[0143] Moreover, these parameters can be calculated from TABLE 9 and TABLE 10 as the following values and satisfy the following conditions:

5th Embodiment			
f (mm)	3.94	T67/T34	1.42
Fno	1.70	T67/T45	2.20
HFOV (deg.)	38.6	T67/T56	7.00
CRA1.0Y (deg.)	34.2	TL/EPD	2.17
V40	3	TL/ImgH	1.55
V30	3	(TL ²)/(EPD × ImgH)	3.36
V20	0	EPD/R14	2.06
T23/T34	0.98	2 × Y72/EPD	2.20
T67/T12	7.00	f/R13 + f/R14	5.68
T67/T23	1.45		

6th Embodiment

[0144] FIG. 11 is a schematic view of an imaging apparatus according to the 6th embodiment of the present disclosure. FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 6th embodiment. In FIG. 11, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 695. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a stop 601, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, a seventh lens element 670, an IR-cut filter 680 and an image surface 690. The image sensor 695 is disposed on the image surface 690 of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (610, 620, 630, 640, 650, 660 and 670) without additional one or more lens elements inserted between the first lens element 610 and the seventh lens element 670.

[0145] The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof. The first lens element 610 is made of a plastic material, and has the object-side surface 611 and the image-side surface 612 being both aspheric.

[0146] The second lens element 620 with negative refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being concave in a paraxial region thereof. The second lens element 620 is made of a plastic material, and has the object-side surface 621 and the image-side surface 622 being both aspheric.

[0147] The third lens element 630 with positive refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave in a paraxial region thereof. The third lens element

630 is made of a plastic material, and has the object-side surface 631 and the image-side surface 632 being both aspheric.

[0148] The fourth lens element 640 with negative refractive power has an object-side surface 641 being concave in a paraxial region thereof and an image-side surface 642 being concave in a paraxial region thereof. The fourth lens element 640 is made of a plastic material, and has the object-side surface 641 and the image-side surface 642 being both aspheric.

[0149] The fifth lens element 650 with positive refractive power has an object-side surface 651 being concave in a paraxial region thereof and an image-side surface 652 being convex in a paraxial region thereof. The fifth lens element 650 is made of a plastic material, and has the object-side surface 651 and the image-side surface 652 being both aspheric.

[0150] The sixth lens element 660 with negative refractive power has an object-side surface 661 being convex in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof. The sixth lens element 660 is made of a plastic material, and has the object-side surface 661 and the image-side surface 662 being both aspheric.

[0151] The seventh lens element 670 with negative refractive power has an object-side surface 671 being convex in a paraxial region thereof and an image-side surface 672 being concave in a paraxial region thereof. The seventh lens element 670 is made of a plastic material, and has the object-side surface 671 and the image-side surface 672 being both aspheric. Moreover, the image-side surface 672 of the seventh lens element 670 includes at least one convex shape in an off-axis region thereof.

[0152] The IR-cut filter 680 is made of a glass material and located between the seventh lens element 670 and the image surface 690, and will not affect the focal length of the image capturing lens assembly.

[0153] The detailed optical data of the 6th embodiment are shown in TABLE 11 and the aspheric surface data are shown in TABLE 12 below.

TABLE 11

6th Embodiment								
f = 3.92 mm, Fno = 1.54, HFOV = 39.0 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.460				
2	Lens 1	1.811	ASP	0.682	Plastic	1.545	56.0	5.28
3		4.242	ASP	0.115				
4	Lens 2	2.917	ASP	0.230	Plastic	1.669	19.5	-15.48
5		2.204	ASP	0.212				
6	Lens 3	3.578	ASP	0.442	Plastic	1.544	56.0	8.51
7		15.059	ASP	-0.153				
8	Stop	Plano		0.384				
9	Lens 4	-52.871	ASP	0.304	Plastic	1.669	19.5	-12.50
10		9.953	ASP	0.106				
11	Lens 5	-3.995	ASP	0.439	Plastic	1.566	37.4	4.16
12		-1.540	ASP	0.069				
13	Lens 6	5.264	ASP	0.341	Plastic	1.669	19.5	-50.93
14		4.441	ASP	0.390				
15	Lens 7	3.079	ASP	0.360	Plastic	1.566	37.4	-4.11
16		1.269	ASP	0.500				

TABLE 11-continued

6th Embodiment							
f = 3.92 mm, Fno = 1.54, HFOV = 39.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
17	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
18		Plano	0.341				
19	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 8 is 1.120 mm.

TABLE 12

Aspheric Coefficients							
Surface #	2	3	4	5	6	7	9
k =	-9.4536E-03	-2.5542E-01	-4.6454E+01	-2.0116E+01	-3.5054E+01	-7.6395E+01	6.9810E+01
A4 =	-2.4715E-03	-1.0957E-01	-3.1358E-02	1.9062E-02	3.2536E-02	-7.9330E-02	-2.5558E-01
A6 =	-5.3775E-04	8.7127E-02	-1.8883E-01	-1.3659E-01	-1.0786E-01	-1.1348E-03	9.0870E-02
A8 =	6.3262E-03	-6.4780E-02	4.6495E-01	2.1178E-01	-1.0401E-02	-1.0639E-01	-1.9343E-01
A10 =	-1.7431E-02	2.6954E-02	-4.8556E-01	-6.7025E-02	1.0937E-01	7.7006E-02	1.4410E-01
A12 =	1.3990E-02	-6.4491E-03	2.6122E-01	-7.8871E-02	-1.5243E-01	-1.6906E-02	2.0349E-01
A14 =	-5.0098E-03	4.0147E-05	-5.6546E-02	6.0261E-02	7.1805E-02		-3.0122E-01
A16 =							1.0090E-01
Surface #	10	11	12	13	14	15	16
k =	-8.9551E+01	8.2993E+00	-1.2392E+00	5.3552E+00	-2.2336E+00	-2.4424E+01	-6.1065E+00
A4 =	-3.2849E-01	-1.8244E-01	1.9668E-01	3.2127E-01	1.9709E-01	-3.3021E-01	-2.3159E-01
A6 =	7.7567E-01	1.2304E+00	-5.1343E-01	-1.0309E+00	-5.6038E-01	2.1461E-01	1.7293E-01
A8 =	-1.9702E+00	-3.0702E+00	9.1278E-01	1.4970E+00	6.6501E-01	-1.8996E-01	-1.1485E-01
A10 =	2.5962E+00	4.0137E+00	-9.5064E-01	-1.4103E+00	-5.1994E-01	1.5512E-01	5.8194E-02
A12 =	-1.7871E+00	-3.0039E+00	5.5090E-01	8.2966E-01	2.7311E-01	-7.5967E-02	-2.0132E-01
A14 =	6.1193E-01	1.2767E+00	-1.6282E-01	-2.8173E-01	-9.4696E-02	2.1820E-02	4.5300E-03
A16 =	-8.2195E-02	-2.8040E-01	1.9125E-02	4.1063E-02	2.0686E-02	-3.6618E-03	-6.3192E-04
A18 =		2.3858E-02		2.2667E-03	-2.5637E-03	3.3407E-04	4.9458E-05
A20 =				-9.7151E-04	1.3673E-04	-1.2833E-05	-1.6540E-06

[0154] In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

[0155] Moreover, these parameters can be calculated from TABLE 11 and TABLE 12 as the following values and satisfy the following conditions:

6th Embodiment			
f (mm)	3.92	T67/T34	1.69
Fno	1.54	T67/T45	3.68
HFOV (deg.)	39.0	T67/T56	5.65
CRA1.0Y (deg.)	35.5	TL/EPD	1.95
V40	5	TL/ImgH	1.54
V30	3	(TL ²)/(EPD × ImgH)	3.00
V20	3	EPD/R14	2.01
T23/T34	0.92	2 × Y72/EPD	1.90
T67/T12	3.39	f/R13 + f/R14	4.36
T67/T23	1.84		

7th Embodiment

[0156] FIG. 13 is a schematic view of an imaging apparatus according to the 7th embodiment of the present disclosure. FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 7th embodiment. In FIG. 13, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 795. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a stop 701, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, a seventh lens element 770, an IR-cut filter 780 and an image surface 790. The image sensor 795 is disposed on the image surface 790 of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (710, 720, 730, 740, 750, 760 and 770) without additional one or more lens elements inserted between the first lens element 710 and the seventh lens element 770.

[0157] The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof. The first lens element 710 is made of a plastic material, and has the object-side surface 711 and the image-side surface 712 being both aspheric.

[0158] The second lens element **720** with negative refractive power has an object-side surface **721** being convex in a paraxial region thereof and an image-side surface **722** being concave in a paraxial region thereof. The second lens element **720** is made of a plastic material, and has the object-side surface **721** and the image-side surface **722** being both aspheric.

[0159] The third lens element **730** with positive refractive power has an object-side surface **731** being convex in a paraxial region thereof and an image-side surface **732** being convex in a paraxial region thereof. The third lens element **730** is made of a plastic material, and has the object-side surface **731** and the image-side surface **732** being both aspheric.

[0160] The fourth lens element **740** with negative refractive power has an object-side surface **741** being concave in a paraxial region thereof and an image-side surface **742** being concave in a paraxial region thereof. The fourth lens element **740** is made of a plastic material, and has the object-side surface **741** and the image-side surface **742** being both aspheric.

[0161] The fifth lens element **750** with positive refractive power has an object-side surface **751** being concave in a paraxial region thereof and an image-side surface **752** being convex in a paraxial region thereof. The fifth lens element

750 is made of a plastic material, and has the object-side surface **751** and the image-side surface **752** being both aspheric.

[0162] The sixth lens element **760** with positive refractive power has an object-side surface **761** being convex in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof. The sixth lens element **760** is made of a plastic material, and has the object-side surface **761** and the image-side surface **762** being both aspheric.

[0163] The seventh lens element **770** with negative refractive power has an object-side surface **771** being convex in a paraxial region thereof and an image-side surface **772** being concave in a paraxial region thereof. The seventh lens element **770** is made of a plastic material, and has the object-side surface **771** and the image-side surface **772** being both aspheric. Moreover, the image-side surface **772** of the seventh lens element **770** includes at least one convex shape in an off-axis region thereof.

[0164] The IR-cut filter **780** is made of a glass material and located between the seventh lens element **770** and the image surface **790**, and will not affect the focal length of the image capturing lens assembly.

[0165] The detailed optical data of the 7th embodiment are shown in TABLE 13 and the aspheric surface data are shown in TABLE 14 below.

TABLE 13

7th Embodiment								
f = 5.26 mm, Fno = 1.66, HFOV = 38.4 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.549					
2	Lens 1	2.490 ASP	0.473	Plastic	1.545	56.0	10.03	
3		4.268 ASP	0.192					
4	Lens 2	3.047 ASP	0.300	Plastic	1.669	19.5	-13.47	
5		2.187 ASP	0.363					
6	Stop	Plano	-0.268					
7	Lens 3	3.652 ASP	0.619	Plastic	1.544	56.0	6.49	
8		-100.000 ASP	0.474					
9	Lens 4	-30.361 ASP	0.350	Plastic	1.669	19.5	-27.65	
10		47.545 ASP	0.378					
11	Lens 5	-6.062 ASP	0.652	Plastic	1.559	40.4	176.12	
12		-5.931 ASP	0.109					
13	Lens 6	1.982 ASP	0.556	Plastic	1.544	56.0	8.97	
14		3.008 ASP	0.992					
15	Lens 7	4.100 ASP	0.420	Plastic	1.544	56.0	-5.82	
16		1.722 ASP	0.500					
17	IR-cut filter	Plano	0.150	Glass	1.517	64.2	—	
18		Plano	0.380					
19	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 6 is 1.500 mm.

TABLE 14

Aspheric Coefficients								
Surface #	2	3	4	5	7	8	9	
k =	-3.6665E-02	-2.0768E+01	-2.3089E+01	-2.4902E+00	2.7747E+00	-8.6102E+00	9.0000E+01	
A4 =	3.5702E-03	3.9323E-02	6.3123E-02	-1.0852E-02	-9.8187E-03	-2.0115E-02	-7.0242E-02	
A6 =	-3.5164E-03	-2.9198E-02	-1.0602E-01	-3.6550E-02	-7.2500E-03	1.8562E-03	-6.5252E-03	
A8 =	4.4562E-03	2.0848E-02	9.0208E-02	2.4768E-02	-2.1798E-02	-2.4534E-03	-7.0043E-03	
A10 =	-3.7751E-03	-9.5301E-03	-4.4461E-02	-4.4316E-03	2.9012E-02	5.9243E-03	1.5208E-02	
A12 =	1.6498E-03	2.7234E-03	1.1761E-02	-5.2657E-04	-1.1337E-02	-2.9429E-03	-7.8688E-03	

TABLE 14-continued

Aspheric Coefficients							
A14 =	-3.0458E-04	-4.0061E-04	-1.3013E-03	9.0587E-05	1.5512E-03	6.7968E-04	1.5070E-03 -5.5120E-05
Surface #	10	11	12	13	14	15	16
k =	-2.0000E+01	-8.6431E+01	-9.0000E+01	-7.6005E-01	-4.4251E-01	-3.1377E-01	-4.7686E+00
A4 =	-3.7254E-02	3.3623E-03	-1.1478E-01	-6.7567E-02	3.9075E-02	-1.6682E-01	-1.0027E-01
A6 =	-1.6632E-02	2.5823E-04	7.6092E-02	1.7614E-02	-4.5414E-02	6.9617E-02	4.7681E-02
A8 =	6.9005E-03	-6.9411E-03	-4.0333E-02	-1.2968E-02	1.6489E-02	-2.2890E-02	-1.7393E-02
A10 =	1.9780E-03	8.3827E-04	1.5289E-02	4.9635E-03	-3.5247E-03	5.6183E-03	4.4096E-03
A12 =	-1.5998E-03	2.4168E-03	-4.6300E-03	-1.0500E-03	4.4990E-04	-9.1855E-04	-7.6029E-04
A14 =	2.8858E-04	-1.5765E-03	1.3196E-03	1.3305E-04	-2.9538E-05	9.8370E-05	8.6361E-05
A16 =	2.5776E-06	3.9407E-04	-3.4733E-04	-2.0210E-05	3.7661E-07	-7.0258E-06	-6.1078E-06
A18 =		-3.3124E-05	6.1387E-05	3.5728E-06	4.7661E-08	3.2127E-07	2.4209E-07
A20 =		-5.5440E-07	-4.6176E-06	-2.5540E-07	-1.2418E-09	-7.2294E-09	-4.0991E-09

[0166] In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

[0167] Moreover, these parameters can be calculated from TABLE 13 and TABLE 14 as the following values and satisfy the following conditions:

7th Embodiment			
f (mm)	5.26	T67/T34	2.09
Fno	1.66	T67/T45	2.62
HFOV (deg.)	38.4	T67/T56	9.10
CRA1.0Y (deg.)	37.0	TL/EPD	2.10
V40	2	TL/ImgH	1.57
V30	2	(TL ²)/(EPD × ImgH)	3.29
V20	2	EPD/R14	1.84
T23/T34	0.20	2 × Y72/EPD	2.14
T67/T12	5.17	f/R13 + f/R14	4.34
T67/T23	10.44		

8th Embodiment

[0168] FIG. 15 is a schematic view of an imaging apparatus according to the 8th embodiment of the present disclosure. FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 8th embodiment. In FIG. 15, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor **895**. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop **800**, a first lens element **810**, a second lens element **820**, a third lens element **830**, a fourth lens element **840**, a fifth lens element **850**, a sixth lens element **860**, a seventh lens element **870**, an IR-cut filter **880** and an image surface **890**. The image sensor **895** is disposed on the image surface **890** of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (**810**, **820**, **830**, **840**, **850**, **860** and **870**) without additional one or more lens elements inserted between the first lens element **810** and the seventh lens element **870**.

[0169] The first lens element **810** with positive refractive power has an object-side surface **811** being convex in a paraxial region thereof and an image-side surface **812** being concave in a paraxial region thereof. The first lens element **810** is made of a plastic material, and has the object-side surface **811** and the image-side surface **812** being both aspheric.

[0170] The second lens element **820** with negative refractive power has an object-side surface **821** being convex in a paraxial region thereof and an image-side surface **822** being concave in a paraxial region thereof. The second lens element **820** is made of a plastic material, and has the object-side surface **821** and the image-side surface **822** being both aspheric.

[0171] The third lens element **830** with positive refractive power has an object-side surface **831** being convex in a paraxial region thereof and an image-side surface **832** being concave in a paraxial region thereof. The third lens element **830** is made of a plastic material, and has the object-side surface **831** and the image-side surface **832** being both aspheric.

[0172] The fourth lens element **840** with negative refractive power has an object-side surface **841** being concave in a paraxial region thereof and an image-side surface **842** being convex in a paraxial region thereof. The fourth lens element **840** is made of a plastic material, and has the object-side surface **841** and the image-side surface **842** being both aspheric.

[0173] The fifth lens element **850** with positive refractive power has an object-side surface **851** being concave in a paraxial region thereof and an image-side surface **852** being convex in a paraxial region thereof. The fifth lens element **850** is made of a plastic material, and has the object-side surface **851** and the image-side surface **852** being both aspheric.

[0174] The sixth lens element **860** with positive refractive power has an object-side surface **861** being convex in a paraxial region thereof and an image-side surface **862** being concave in a paraxial region thereof. The sixth lens element **860** is made of a plastic material, and has the object-side surface **861** and the image-side surface **862** being both aspheric.

[0175] The seventh lens element **870** with negative refractive power has an object-side surface **871** being convex in a paraxial region thereof and an image-side surface **872** being concave in a paraxial region thereof. The seventh lens

element **870** is made of a plastic material, and has the object-side surface **871** and the image-side surface **872** being both aspheric. Moreover, the image-side surface **872** of the seventh lens element **870** includes at least one convex shape in an off-axis region thereof.

[0176] The IR-cut filter **880** is made of a glass material and located between the seventh lens element **870** and the image surface **890**, and will not affect the focal length of the image capturing lens assembly.

[0177] The detailed optical data of the 8th embodiment are shown in TABLE 15 and the aspheric surface data are shown in TABLE 16 below.

[0178] In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

[0179] Moreover, these parameters can be calculated from TABLE 15 and TABLE 16 as the following values and satisfy the following conditions:

TABLE 15

8th Embodiment							
f = 5.32 mm, Fno = 1.55, HFOV = 38.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.542				
2	Lens 1	2.708 ASP	0.573	Plastic	1.545	56.1	8.64
3		5.895 ASP	0.077				
4	Lens 2	3.244 ASP	0.300	Plastic	1.650	21.5	-9.10
5		2.019 ASP	0.080				
6	Lens 3	3.062 ASP	0.764	Plastic	1.544	56.0	6.02
7		43.137 ASP	0.438				
8	Lens 4	-31.569 ASP	0.353	Plastic	1.639	23.5	-72.38
9		-100.000 ASP	0.301				
10	Lens 5	-5.230 ASP	0.819	Plastic	1.544	56.0	177.98
11		-5.236 ASP	0.050				
12	Lens 6	2.028 ASP	0.521	Plastic	1.534	55.9	10.35
13		2.918 ASP	1.044				
14	Lens 7	5.843 ASP	0.427	Plastic	1.534	55.9	-5.71
15		1.953 ASP	0.500				
16	IR-cut filter	Plano	0.150	Glass	1.517	64.2	—
17		Plano	0.356				
18	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 15 is 3.400 mm.

TABLE 16

Aspheric Coefficients							
Surface #	2	3	4	5	6	7	8
k =	-4.9818E-01	-2.0768E+01	-2.5344E+01	-2.5045E+00	1.8024E+00	-8.6102E+00	9.0000E+01
A4 =	4.7203E-03	5.3473E-02	8.9287E-02	-1.1999E-03	-1.2801E-02	-1.7803E-02	-6.0269E-02
A6 =	-5.3729E-04	-6.8352E-02	-1.4462E-01	-4.0489E-02	4.2010E-03	1.6589E-03	-1.5721E-02
A8 =	-8.7590E-04	4.5120E-02	1.0661E-01	1.7591E-02	-3.2793E-02	1.8893E-03	6.1518E-03
A10 =	6.7349E-05	-1.7186E-02	-4.4467E-02	4.9188E-04	2.9161E-02	3.9254E-04	4.4800E-03
A12 =	1.2547E-04	3.5777E-03	9.9435E-03	-1.6417E-03	-9.4901E-03	-2.8986E-04	-2.5870E-03
A14 =	-4.4185E-05	-3.3340E-04	-9.3018E-04	2.2665E-04	1.1106E-03	1.4827E-04	5.9455E-04
A16 =							-7.3364E-05
Surface #	9	10	11	12	13	14	15
k =	-2.0000E+01	-8.6431E+01	-9.0000E+01	-8.5023E-01	-3.9054E-01	7.9729E-01	-5.4844E+00
A4 =	-2.3402E-02	-1.2625E-02	-1.2551E-01	-5.6841E-02	3.5271E-02	-1.4707E-01	-9.0133E-02
A6 =	-2.6129E-02	3.3613E-02	1.0560E-01	1.3480E-02	-4.2435E-02	5.3927E-02	4.0519E-02
A8 =	8.1134E-03	-7.5950E-02	-8.8913E-02	-1.4428E-02	1.3837E-02	-1.4235E-02	-1.4448E-02
A10 =	4.3615E-03	7.9621E-02	5.7560E-02	7.5632E-03	-2.2591E-03	2.1354E-03	3.6220E-03
A12 =	-2.5727E-03	-5.6998E-02	-2.7448E-02	-2.6655E-03	8.2393E-05	7.0196E-05	-6.2213E-04
A14 =	5.1744E-04	2.8098E-02	8.9776E-03	7.1040E-04	3.8541E-05	-8.1365E-05	7.0950E-05
A16 =	-2.7450E-05	-8.9132E-03	-1.8692E-03	-1.4304E-04	-7.4045E-06	1.2385E-05	-5.0613E-06
A18 =		1.6058E-03	2.1902E-04	1.7795E-05	5.3504E-07	-8.0796E-07	2.0257E-07
A20 =		-1.2272E-04	-1.0729E-05	-9.3503E-07	-1.3730E-08	2.0017E-08	-3.4591E-09

8th Embodiment			
f (mm)	5.32	T67/T34	2.38
Fno	1.55	T67/T45	3.47
HFOV (deg.)	38.0	T67/T56	20.88
CRA1.0Y (deg.)	38.1	TL/EPD	1.97
V40	2	TL/ImgH	1.59
V30	2	(TL ²)/(EPD × ImgH)	3.14
V20	0	EPD/R14	1.76
T23/T34	0.18	2 × Y72/EPD	1.98
T67/T12	13.56	f/R13 + f/R14	3.63
T67/T23	13.05		

9th Embodiment

[0180] FIG. 17 is a schematic view of an imaging apparatus according to the 9th embodiment of the present disclosure. FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 9th embodiment. In FIG. 17, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor 995. The image capturing lens assembly includes, in order from an object side to an image side, a stop 901, a first lens element 910, a second lens element 920, an aperture stop 900, a third lens element 930, a fourth lens element 940, a fifth lens element 950, a sixth lens element 960, a seventh lens element 970, an IR-cut filter 980 and an image surface 990. The image sensor 995 is disposed on the image surface 990 of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (910, 920, 930, 940, 950, 960 and 970) without additional one or more lens elements inserted between the first lens element 910 and the seventh lens element 970.

[0181] The first lens element 910 with positive refractive power has an object-side surface 911 being convex in a paraxial region thereof and an image-side surface 912 being concave in a paraxial region thereof. The first lens element 910 is made of a plastic material, and has the object-side surface 911 and the image-side surface 912 being both aspheric.

[0182] The second lens element 920 with negative refractive power has an object-side surface 921 being convex in a paraxial region thereof and an image-side surface 922 being concave in a paraxial region thereof. The second lens

element 920 is made of a plastic material, and has the object-side surface 921 and the image-side surface 922 being both aspheric.

[0183] The third lens element 930 with positive refractive power has an object-side surface 931 being convex in a paraxial region thereof and an image-side surface 932 being concave in a paraxial region thereof. The third lens element 930 is made of a plastic material, and has the object-side surface 931 and the image-side surface 932 being both aspheric.

[0184] The fourth lens element 940 with positive refractive power has an object-side surface 941 being convex in a paraxial region thereof and an image-side surface 942 being concave in a paraxial region thereof. The fourth lens element 940 is made of a plastic material, and has the object-side surface 941 and the image-side surface 942 being both aspheric.

[0185] The fifth lens element 950 with positive refractive power has an object-side surface 951 being concave in a paraxial region thereof and an image-side surface 952 being convex in a paraxial region thereof. The fifth lens element 950 is made of a plastic material, and has the object-side surface 951 and the image-side surface 952 being both aspheric.

[0186] The sixth lens element 960 with positive refractive power has an object-side surface 961 being convex in a paraxial region thereof and an image-side surface 962 being concave in a paraxial region thereof. The sixth lens element 960 is made of a plastic material, and has the object-side surface 961 and the image-side surface 962 being both aspheric.

[0187] The seventh lens element 970 with negative refractive power has an object-side surface 971 being convex in a paraxial region thereof and an image-side surface 972 being concave in a paraxial region thereof. The seventh lens element 970 is made of a plastic material, and has the object-side surface 971 and the image-side surface 972 being both aspheric. Moreover, the image-side surface 972 of the seventh lens element 970 includes at least one convex shape in an off-axis region thereof.

[0188] The IR-cut filter 980 is made of a glass material and located between the seventh lens element 970 and the image surface 990, and will not affect the focal length of the image capturing lens assembly.

[0189] The detailed optical data of the 9th embodiment are shown in TABLE 17 and the aspheric surface data are shown in TABLE 18 below.

TABLE 17

9th Embodiment								
f = 5.38 mm, Fno = 1.64, HFOV = 37.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #		Focal Length
0	Object	Plano	Infinity					
1	Stop	Plano	-0.508					
2	Lens 1	2.637 ASP	0.596	Plastic	1.545	56.1		8.52
3		5.617 ASP	0.090					
4	Lens 2	3.468 ASP	0.300	Plastic	1.650	21.5		-8.77
5		2.083 ASP	0.369					
6	Ape. Stop	Plano	-0.292					
7	Lens 3	3.062 ASP	0.666	Plastic	1.544	56.0		6.37
8		24.259 ASP	0.414					
9	Lens 4	19.837 ASP	0.350	Plastic	1.639	23.5		151.19
10		24.794 ASP	0.332					
11	Lens 5	-4.841 ASP	0.841	Plastic	1.544	56.0		52.07
12		-4.388 ASP	0.085					

TABLE 17-continued

9th Embodiment f = 5.38 mm, Fno = 1.64, HFOV = 37.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
13	Lens 6	2.229	ASP	0.533	Plastic	1.534	55.9	13.82
14		2.928	ASP	1.054				
15	Lens 7	6.474	ASP	0.420	Plastic	1.534	55.9	-5.60
16		1.999	ASP	0.500				
17	IR-cut filter	Plano	0.150	Glass	1.517	64.2		—
18		Plano	0.345					
19	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 1 is 1.650 mm.

TABLE 18

Aspheric Coefficients							
Surface #	2	3	4	5	7	8	9
k =	-4.5445E-01	-2.1761E+01	-2.6140E+01	-2.4105E+00	1.8753E+00	-9.0000E+01	-7.6075E+01
A4 =	4.1302E-03	3.2581E-02	5.1941E-02	-2.6960E-02	-3.0479E-02	-1.9186E-02	-5.1391E-02
A6 =	-2.7932E-03	-2.8682E-02	-7.4595E-02	1.5222E-02	3.0393E-02	-2.5446E-03	-2.3208E-02
A8 =	2.7958E-03	1.4736E-02	4.9254E-02	-3.0722E-02	-4.8862E-02	9.8239E-03	1.0201E-02
A10 =	-2.2753E-03	-5.2315E-03	-1.9965E-02	2.3762E-02	3.5506E-02	-5.9129E-03	-1.7966E-03
A12 =	8.3060E-04	1.2253E-03	4.6585E-03	-7.8795E-03	-1.1156E-02	2.6979E-03	1.9915E-03
A14 =	-1.3085E-04	-1.6143E-04	-4.7812E-04	9.3533E-04	1.3164E-03	-3.9240E-04	-7.2069E-04
A16 =							4.5597E-05
Surface #	10	11	12	13	14	15	16
k =	6.4983E+01	-7.9970E+01	-7.6733E+01	-7.5882E-01	-5.2630E-01	1.0878E+00	-7.3030E+00
A4 =	-1.5669E-02	-3.4595E-02	-1.4314E-01	-5.2624E-02	2.1822E-02	-1.4907E-01	-7.6094E-02
A6 =	-3.1011E-02	7.2904E-02	1.4516E-01	9.9099E-03	-3.2152E-02	6.1702E-02	3.2174E-02
A8 =	1.1306E-02	-1.1773E-01	-1.3910E-01	-1.2308E-02	1.0227E-02	-1.9662E-02	-1.0963E-02
A10 =	-4.0888E-04	1.1392E-01	9.9086E-02	7.0215E-03	-1.6518E-03	4.4931E-03	2.6426E-03
A12 =	8.0736E-05	-7.8718E-02	-5.0083E-02	-2.7622E-03	8.6889E-05	-6.2296E-04	-4.4229E-04
A14 =	-3.2303E-05	3.7755E-02	1.7030E-02	8.0529E-04	1.5968E-05	4.7248E-05	4.9607E-05
A16 =	7.0949E-06	-1.1702E-02	-3.6714E-03	-1.6795E-04	-3.1585E-06	-1.7668E-06	-3.4892E-06
A18 =		2.0850E-03	4.4987E-04	2.0937E-05	1.9361E-07	3.2716E-08	1.3750E-07
A20 =		-1.6015E-04	-2.3624E-05	-1.0968E-06	-3.2506E-09	-7.6015E-10	-2.3066E-09

[0190] In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

[0191] Moreover, these parameters can be calculated from TABLE 17 and TABLE 18 as the following values and satisfy the following conditions:

9th Embodiment			
f (mm)	5.38	T67/T34	2.55
Fno	1.64	T67/T45	3.17
HFOV (deg.)	37.9	T67/T56	12.40
CRA1.0Y (deg.)	42.2	TL/EPD	2.06
V40	2	TL/ImgH	1.59
V30	2	(TL ²)/(EPD × ImgH)	3.28
V20	0	EPD/R14	1.64
T23/T34	0.19	2 × Y72/EPD	2.05

-continued

9th Embodiment			
T67/T12	11.71	f/R13 + f/R14	3.52
T67/T23	13.69		

10th Embodiment

[0192] FIG. 19 is a schematic view of an imaging apparatus according to the 10th embodiment of the present disclosure. FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging apparatus according to the 10th embodiment. In FIG. 19, the imaging apparatus includes an image capturing lens assembly (its reference numeral is omitted) and an image sensor **1095**. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop **1000**, a first lens element **1010**, a second lens element **1020**, a stop **1001**, a third lens element **1030**, a fourth lens element **1040**, a fifth lens element **1050**, a sixth lens element **1060**, a seventh lens element **1070**, an IR-cut filter **1080** and an image surface **1090**. The image sensor **1095** is disposed on

the image surface **1090** of the image capturing lens assembly. The image capturing lens assembly includes seven lens elements (**1010**, **1020**, **1030**, **1040**, **1050**, **1060** and **1070**) without additional one or more lens elements inserted between the first lens element **1010** and the seventh lens element **1070**.

[0193] The first lens element **1010** with positive refractive power has an object-side surface **1011** being convex in a paraxial region thereof and an image-side surface **1012** being concave in a paraxial region thereof. The first lens element **1010** is made of a plastic material, and has the object-side surface **1011** and the image-side surface **1012** being both aspheric.

[0194] The second lens element **1020** with negative refractive power has an object-side surface **1021** being convex in a paraxial region thereof and an image-side surface **1022** being concave in a paraxial region thereof. The second lens element **1020** is made of a plastic material, and has the object-side surface **1021** and the image-side surface **1022** being both aspheric.

[0195] The third lens element **1030** with positive refractive power has an object-side surface **1031** being convex in a paraxial region thereof and an image-side surface **1032** being concave in a paraxial region thereof. The third lens element **1030** is made of a plastic material, and has the object-side surface **1031** and the image-side surface **1032** being both aspheric.

[0196] The fourth lens element **1040** with negative refractive power has an object-side surface **1041** being convex in a paraxial region thereof and an image-side surface **1042** being concave in a paraxial region thereof. The fourth lens

element **1040** is made of a plastic material, and has the object-side surface **1041** and the image-side surface **1042** being both aspheric.

[0197] The fifth lens element **1050** with negative refractive power has an object-side surface **1051** being concave in a paraxial region thereof and an image-side surface **1052** being convex in a paraxial region thereof. The fifth lens element **1050** is made of a plastic material, and has the object-side surface **1051** and the image-side surface **1052** being both aspheric.

[0198] The sixth lens element **1060** with positive refractive power has an object-side surface **1061** being convex in a paraxial region thereof and an image-side surface **1062** being concave in a paraxial region thereof. The sixth lens element **1060** is made of a plastic material, and has the object-side surface **1061** and the image-side surface **1062** being both aspheric.

[0199] The seventh lens element **1070** with negative refractive power has an object-side surface **1071** being convex in a paraxial region thereof and an image-side surface **1072** being concave in a paraxial region thereof. The seventh lens element **1070** is made of a plastic material, and has the object-side surface **1071** and the image-side surface **1072** being both aspheric.

[0200] Moreover, the image-side surface **1072** of the seventh lens element **1070** includes at least one convex shape in an off-axis region thereof.

[0201] The IR-cut filter **1080** is made of a glass material and located between the seventh lens element **1070** and the image surface **1090**, and will not affect the focal length of the image capturing lens assembly.

[0202] The detailed optical data of the 10th embodiment are shown in TABLE 19 and the aspheric surface data are shown in TABLE 20 below.

TABLE 19

10th Embodiment							
f = 5.33 mm, Fno = 1.68, HFOV = 38.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.386				
2	Lens 1	2.709 ASP	0.491	Plastic	1.545	56.0	8.74
3		5.891 ASP	0.089				
4	Lens 2	2.531 ASP	0.300	Plastic	1.669	19.5	-12.08
5		1.836 ASP	0.476				
6	Stop	Plano	-0.346				
7	Lens 3	3.144 ASP	0.630	Plastic	1.544	56.0	7.08
8		15.950 ASP	0.657				
9	Lens 4	55.808 ASP	0.357	Plastic	1.669	19.5	-14.33
10		8.157 ASP	0.250				
11	Lens 5	-3.487 ASP	0.562	Plastic	1.584	28.2	-108.11
12		-3.910 ASP	0.030				
13	Lens 6	1.764 ASP	0.441	Plastic	1.544	56.0	7.58
14		2.812 ASP	0.819				
15	Lens 7	1.754 ASP	0.420	Plastic	1.534	55.9	-11.54
16		1.252 ASP	0.900				
17	IR-cut filter	Plano	0.150	Glass	1.517	64.2	—
18		Plano	0.416				
19	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

The effective radius of the surface 6 is 1.500 mm.

The effective radius of the surface 10 is 1.700 mm.

TABLE 20

Aspheric Coefficients							
Surface #	2	3	4	5	7	8	9
k =	-5.7496E-01	-2.8958E+01	-1.3404E+01	-2.0233E+00	2.1586E+00	4.8599E+01	5.5528E+01
A4 =	5.5780E-03	4.9851E-02	8.7403E-02	-1.6264E-02	-2.2005E-02	-1.6334E-02	-9.2506E-02
A6 =	-6.4312E-04	-7.4482E-02	-1.3685E-01	-2.5899E-02	1.1079E-02	1.0346E-02	2.7995E-02
A8 =	-6.0604E-03	6.0867E-02	1.1531E-01	1.6193E-02	-3.5614E-02	-1.1716E-02	-9.2854E-02
A10 =	4.8065E-03	-2.9352E-02	-5.4439E-02	2.6964E-03	3.5734E-02	1.2123E-02	1.0164E-01
A12 =	-2.0278E-03	6.9792E-03	1.3415E-02	-3.6892E-03	-1.3349E-02	-5.1207E-03	-5.0803E-02
A14 =	2.7313E-04	-6.5355E-04	-1.3389E-03	6.0768E-04	1.7721E-03	9.9384E-04	1.2239E-02
A16 =							-1.1486E-03
Surface #	10	11	12	13	14	15	16
k =	-6.1099E+01	-7.3167E+01	-6.7479E+01	-7.1519E-01	-7.4585E-01	-1.5084E+00	-3.4812E+00
A4 =	-3.7724E-02	-9.5964E-02	-2.1191E-01	-7.7249E-02	6.4199E-02	-2.1939E-01	-1.1960E-01
A6 =	3.6230E-02	2.3555E-01	2.0822E-01	4.4998E-02	-6.0101E-02	1.0137E-01	5.5377E-02
A8 =	-6.2840E-02	-2.4185E-01	-1.5253E-01	-3.9850E-02	1.9908E-02	-3.6815E-02	-2.0065E-02
A10 =	3.8952E-02	1.4468E-01	8.6808E-02	1.8602E-02	-3.8127E-03	1.0069E-02	5.1213E-03
A12 =	-1.0683E-02	-5.8680E-02	-3.6499E-02	-5.1546E-03	4.3619E-04	-1.8733E-03	-8.8807E-04
A14 =	1.0668E-03	1.7638E-02	1.0716E-02	8.6017E-04	-2.7805E-05	2.2575E-04	1.0122E-04
A16 =	1.7755E-05	-3.8990E-03	-2.0085E-03	-8.2530E-05	8.0902E-07	-1.6844E-05	-7.1724E-06
A18 =		5.5235E-04	2.1081E-04	4.0513E-06	-1.4927E-08	7.0697E-07	2.8417E-07
A20 =		-3.6177E-05	-9.3392E-06	-7.3815E-08	9.7637E-10	-1.2752E-08	-4.7902E-09

[0203] In the 10th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. A4-A20 represent the aspheric coefficients ranging from the 4th order to the 20th order. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment, so an explanation in this regard will not be provided again.

[0204] Moreover, these parameters can be calculated from TABLE 19 and TABLE 20 as the following values and satisfy the following conditions:

10th Embodiment			
f (mm)	5.33	T67/T34	1.25
Fno	1.68	T67/T45	3.28
HFOV (deg.)	38.0	T67/T56	27.30
CRA1.0Y (deg.)	30.6	TL/EPD	2.09
V40	3	TL/ImgH	1.57
V30	3	(TL ²)/(EPD × ImgH)	3.28
V20	2	EPD/R14	2.53
T23/T34	0.20	2 × Y72/EPD	2.15
T67/T12	9.20	f/R13 + f/R14	7.30
T67/T23	6.30		

11th Embodiment

[0205] FIG. 23 is a three-dimensional view of an imaging apparatus 10 according to the 11th embodiment of the present disclosure. In FIG. 23, the imaging apparatus 10 according to the 11th embodiment is a camera module. The imaging apparatus 10 includes an imaging lens module 11, a driving assembly 12 and an image sensor 13, wherein the imaging lens module 11 includes the image capturing lens assembly according to the 1st embodiment and a barrel (its reference numeral is omitted) for carrying the image capturing lens assembly. An image of an imaged object can be captured by the imaging apparatus 10 via the imaging lens module 11, the driving assembly 12 is used to bring the

image into focus so that the image can be clearly formed on the image sensor 13, and then the data of the image is outputted.

[0206] The driving assembly 12 can have an auto-focus functionality, and a driving method thereof can use a voice coil motor (VCM), a micro electro-mechanical system (MEMS), a piezoelectric system or a shape memory alloy system. The driving assembly 12 enables the image capturing lens assembly to obtain a preferable imaging position, so that the imaged object in different object distances can be imaged clearly.

[0207] The image sensor 13 of the imaging apparatus 10 can have the properties of high photosensitivity and low noise (such as CMOS and CCD) and is disposed on the image surface of the image capturing lens assembly, so that the excellent image quality of the image capturing lens assembly can be truly presented.

[0208] Moreover, the imaging apparatus 10 can further include an image stabilizing module 14. The image stabilizing module 14 can exemplarily include an accelerator, a gyroscope or a Hall Effect sensor. In the 11th embodiment, the image stabilizing module 14 is a gyroscope. However, it is only exemplary and the image stabilizing module 14 is not limited thereto. By adjusting the changes in different axial directions of the image capturing lens assembly, the blurry image due to motion during exposure can be compensated, so that the image quality of dynamic scenes or low-light scenes can be enhanced. Moreover, the advanced image compensation functions, such as optical image stabilization (OIS) or electronic image stabilization (EIS), can be provided.

12th Embodiment

[0209] FIG. 24A is a schematic view showing a side of an electronic device 20 according to the 12th embodiment of the present disclosure. FIG. 24B is a schematic view showing another side of the electronic device 20 in FIG. 24A. FIG. 24C is a block diagram of the electronic device 20 in FIG. 24A. In FIG. 24A, FIG. 24B and FIG. 24C, the electronic device 20 of the 12th embodiment is a smart-

phone. The electronic device 20 includes the imaging apparatus 10, a flash module 21, an auxiliary focusing module 22, an image signal processor 23, a user interface 24 and an image software processor 25. When a user takes a photograph via the user interface 24, light rays of the imaged object 26 are focused by the electronic device 20 via the imaging apparatus 10 for generating an image. Meanwhile, a light compensation function is provided by the flash module 21, the object distance of the imaged object 26 is provided by the auxiliary focusing module 22 for quick focusing, and an optimized image processing is provided by the image signal processor 23 and the image software processor 25, so that the image quality of the image capturing lens assembly can be further enhanced. The auxiliary focusing module 22 can adopt conventional infrared or laser for quick focusing. The user interface 24 can adopt a touch screen or a physical button, and an image software processor 25 can be coordinated with the user interface 24 for providing a variety of shooting modes and a variety of image processing modifications.

[0210] The imaging apparatus 10 of the 12th embodiment can be the same as the imaging apparatus 10 of the 11th embodiment, and are not repeated herein.

13th Embodiment

[0211] FIG. 25 is a schematic view of an electronic device 30 according to the 13th embodiment of the present disclosure. The electronic device 30 of the 13th embodiment is a tablet personal computer. The electronic device 30 includes an imaging apparatus 31. The imaging apparatus 31 can be the same as that of the 11th embodiment, and will not be repeated herein.

14th Embodiment

[0212] FIG. 26 is a schematic view of an electronic device 40 according to the 14th embodiment of the present disclosure. The electronic device 40 of the 14th embodiment is a wearable device. The electronic device 40 includes an imaging apparatus 41. The imaging apparatus 41 can be the same as that of the 11th embodiment, and will not be repeated herein.

[0213] The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-20 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An image capturing lens assembly comprising seven lens elements, the seven lens elements being, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element;

wherein the fifth lens element with positive refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof; the sixth lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, and the object-side surface and the image-side surface of the sixth lens element are aspheric; the seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, the image-side surface of the seventh lens element comprises at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the seventh lens element are aspheric; and

wherein an axial distance between an object-side surface of the first lens element and an image surface is TL, an entrance pupil diameter of the image capturing lens assembly is EPD, and the following condition is satisfied:

$$1.0 < TL/EPD \leq 2.40.$$

2. The image capturing lens assembly of claim 1, wherein an object-side surface of the fourth lens element is concave in a paraxial region thereof.

3. The image capturing lens assembly of claim 1, wherein the axial distance between the object-side surface of the first lens element and the image surface is TL, a maximum image height of the image capturing lens assembly is ImgH, an f-number of the image capturing lens assembly is Fno, and the following conditions are satisfied:

$$0.80 < TL/ImgH < 1.70; \text{ and}$$

$$1.0 < Fno \leq 1.70.$$

4. The image capturing lens assembly of claim 1, wherein a vertical distance between a maximum effective diameter position on the image-side surface of the seventh lens element and an optical axis is Y72, the entrance pupil diameter of the image capturing lens assembly is EPD, and the following condition is satisfied:

$$1.30 < 2 \times Y72/EPD < 2.30.$$

5. The image capturing lens assembly of claim 1, wherein the first lens element with positive refractive power has the object-side surface being convex in a paraxial region thereof, an incident angle of a chief ray at a maximum image height on the image surface of the image capturing lens assembly is CRA1.0Y, and the following condition is satisfied:

$$32.0 \text{ degrees} \leq CRA1.0Y.$$

6. The image capturing lens assembly of claim 1, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, an axial distance between the sixth lens element and the seventh lens element is T67, and the following conditions are satisfied:

$$1.0 < T67/T12;$$

$$1.0 < T67/T23;$$

$$1.0 < T67/T34;$$

$$1.0 < T67/T45; \text{ and}$$

$$1.0 < T67/T56.$$

7. The image capturing lens assembly of claim 1, wherein an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, and the following condition is satisfied:

$$0 < T23/T34 \leq 0.90.$$

8. The image capturing lens assembly of claim 1, wherein a focal length of the image capturing lens assembly is f, a curvature radius of the object-side surface of the seventh lens element is R13, a curvature radius of the image-side surface of the seventh lens element is R14, and the following condition is satisfied:

$$4.0 < (f/R13) + (f/R14) < 10.0.$$

9. The image capturing lens assembly of claim 1, wherein the axial distance between the object-side surface of the first lens element and the image surface is TL, the entrance pupil diameter of the image capturing lens assembly is EPD, and the following condition is satisfied:

$$1.30 < TL/EPD \leq 2.0.$$

10. The image capturing lens assembly of claim 1, wherein the entrance pupil diameter of the image capturing lens assembly is EPD, a curvature radius of the image-side surface of the seventh lens element is R14, and the following condition is satisfied:

$$1.50 < EPD/R14 < 4.0.$$

11. The image capturing lens assembly of claim 1, wherein the axial distance between the object-side surface of the first lens element and the image surface is TL, the entrance pupil diameter of the image capturing lens assembly is EPD, a maximum image height of the image capturing lens assembly is ImgH, and the following condition is satisfied:

$$2.0 < (TL^2)/(EPD \times \text{ImgH}) < 3.20.$$

12. The image capturing lens assembly of claim 11, wherein the axial distance between the object-side surface of the first lens element and the image surface is TL, the entrance pupil diameter of the image capturing lens assembly is EPD, the maximum image height of the image capturing lens assembly is ImgH, and the following condition is satisfied:

$$2.0 < (TL^2)/(EPD \times \text{ImgH}) \leq 3.0.$$

13. The image capturing lens assembly of claim 1, wherein a total number of lens elements having Abbe numbers less than 40 is V40, and the following condition is satisfied:

$$4 \leq V40.$$

14. The image capturing lens assembly of claim 1, wherein a total number of lens elements having Abbe numbers less than 30 is V30, and the following condition is satisfied:

$$3 \leq V30.$$

15. The image capturing lens assembly of claim 1, wherein a total number of lens elements having Abbe numbers less than 20 is V20, and the following condition is satisfied:

$$2 \leq V20.$$

16. An imaging apparatus, comprising:

the image capturing lens assembly of claim 1; and

an image sensor, wherein the image sensor is disposed on the image surface of the image capturing lens assembly.

17. An electronic device, comprising:

the imaging apparatus of claim 16.

18. An image capturing lens assembly comprising seven lens elements, the seven lens elements being, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element;

wherein the fifth lens element has positive refractive power; the sixth lens element has an image-side surface being concave in a paraxial region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric; the seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, the image-side surface of the seventh lens element comprises at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the seventh lens element are aspheric; and

wherein an axial distance between an object-side surface of the first lens element and an image surface is TL, an entrance pupil diameter of the image capturing lens assembly is EPD, and the following condition is satisfied:

$$1.20 < TL/EPD \leq 2.10.$$

19. The image capturing lens assembly of claim 18, wherein an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, and the following condition is satisfied:

$$0 < T23/T34 \leq 0.90.$$

20. The image capturing lens assembly of claim 18, wherein a vertical distance between a maximum effective diameter position on the image-side surface of the seventh lens element and an optical axis is Y72, the entrance pupil diameter of the image capturing lens assembly is EPD, and the following condition is satisfied:

$$1.30 < 2 \times Y72/EPD < 2.30.$$

21. The image capturing lens assembly of claim 18, wherein the entrance pupil diameter of the image capturing lens assembly is EPD, a curvature radius of the image-side surface of the seventh lens element is R14, and the following condition is satisfied:

$$1.5 < EPD/R14 < 4.0.$$

22. The image capturing lens assembly of claim 18, wherein the axial distance between the object-side surface of the first lens element and the image surface is TL, the entrance pupil diameter of the image capturing lens assembly is EPD, and the following condition is satisfied:

bly is EPD, a maximum image height of the image capturing lens assembly is ImgH, and the following condition is satisfied:

$$2.0 < (TL^2)/(EPD \times \text{ImgH}) < 3.20.$$

23. The image capturing lens assembly of claim **18**, wherein an object-side surface of the fourth lens element is concave in a paraxial region thereof.

24. The image capturing lens assembly of claim **18**, wherein a focal length of the image capturing lens assembly is f , a curvature radius of the object-side surface of the seventh lens element is R13, a curvature radius of the image-side surface of the seventh lens element is R14, and the following condition is satisfied:

$$4.0 < (f/R13) + (f/R14) < 10.0.$$

25. The image capturing lens assembly of claim **18**, wherein a total number of lens elements having Abbe numbers less than 40 is V40, and the following condition is satisfied:

$$4 \leq V40.$$

26. The image capturing lens assembly of claim **18**, wherein a total number of lens elements having Abbe numbers less than 20 is V20, and the following condition is satisfied:

$$2.0 \leq V20.$$

27. An image capturing lens assembly comprising seven lens elements, the seven lens elements being, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, a sixth lens element and a seventh lens element;

wherein the sixth lens element has an image-side surface being concave in a paraxial region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric; the seventh lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, the image-side surface of the seventh lens element comprises at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the seventh lens element are aspheric; and

wherein an axial distance between an object-side surface of the first lens element and an image surface is TL, an entrance pupil diameter of the image capturing lens assembly is EPD, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, and the following conditions are satisfied:

$$1.0 < TL/EPD \leq 2.10; \text{ and}$$

$$0 < T23/T34 \leq 0.90.$$

28. The image capturing lens assembly of claim **27**, wherein an object-side surface of the fourth lens element is concave in a paraxial region thereof.

29. The image capturing lens assembly of claim **27**, wherein a focal length of the image capturing lens assembly is f , a curvature radius of the object-side surface of the seventh lens element is R13, a curvature radius of the image-side surface of the seventh lens element is R14, and the following condition is satisfied:

$$4.0 < (f/R13) + (f/R14) < 10.0.$$

30. The image capturing lens assembly of claim **27**, wherein the axial distance between the object-side surface of the first lens element and the image surface is TL, the entrance pupil diameter of the image capturing lens assembly is EPD, a maximum image height of the image capturing lens assembly is ImgH, and the following condition is satisfied:

$$2.0 < (TL^2)/(EPD \times \text{ImgH}) < 3.20.$$

31. The image capturing lens assembly of claim **27**, wherein a total number of lens elements having Abbe numbers less than 40 is V40, and the following condition is satisfied:

$$4 \leq V40.$$

32. The image capturing lens assembly of claim **27**, wherein a total number of lens elements having Abbe numbers less than 20 is V20, and the following condition is satisfied:

$$2 \leq V20.$$

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